Bachelor's Thesis at Lucerne University of Applied Sciences and Arts - Technology & Architecture

Title **Hydraulic Integration of a Circulation Heat Pump into the Cold and Hot Water Network**

Student: Tobias Eichelberger

Bachelor's Program **Bachelor's in Building Services**

Engineering | Energy Graduate: Valerio Pietro Paganini

Bachelor's Program **Bachelor's in Building Services**

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Instructor **Von Euw, Reto | Zehnder,**

Timotheus Expert **Hess, Roni**

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authoritative.

Abstract (German)

Rising cold water temperatures in buildings can lead to reduced comfort and hygienically unfavorable operating conditions. At the same time, continuous circulation heat losses occur in water heating systems. The aim of this bachelor’s thesis is to investigate the hydraulic integration of a circulation heat pump into the cold- and hot-water networks and to evaluate its potential for utilizing heat from the cold-water network.

The methodological basis consists of theoretical analyses, energy balances, the integration of two laboratory setups, the commissioning of the test bench, a qualitative evaluation of measurements, and a simplified calculation model. In addition, control strategies for different temperature conditions, design aspects for new and existing buildings, and requirements for a future market-ready circulation heat pump are developed.

The calculations show a heat loss of 484.5 W from the hot water distribution system in the representative apartment building. This is offset by a potential heat uptake of 329.7 W from the cold water network. The usable potential is highly location-dependent. While alpine regions have only limited potential due to low cold-water temperatures, urban areas with higher summer temperatures may be more suitable. The hydraulic function was qualitatively confirmed on the test bench. The heat pump extracted heat from the cold-water system and transferred it to the upper section of the hot-water storage tank. The stratification of the storage tank was largely maintained. However, due to a compressor failure, it was not possible to conduct a complete series of measurements.

The calculation model shows that the circulation heat pump can provide energy support to the hot water storage tank. In the model, 4,380 kWh/a of heat is transferred to the storage tank, of which 2,947 kWh/a is extracted from the cold water system. This results in a theoretical annual coefficient of performance of 3.06. The main charge of 11,066 kWh/a remains necessary. For future implementation, a hydraulically decoupled integration, stable control, SVGW-compliant drinking water components, electrical power measurements, and further reproducible measurement series are necessary.

Abstract (English)

Rising cold water temperatures can reduce comfort and create hygienically unfavorable operating conditions in buildings. At the same time, domestic hot water systems experience continuous circulation heat losses. This thesis investigates the hydraulic integration of a circulation heat pump into a cold and hot water system and assesses its potential for utilizing heat from the cold water network.

The methodology is based on theoretical analyses, energy balances, the combination of two laboratory test rigs, commissioning, qualitative test measurements, and a simplified calculation model. In addition, control strategies, planning aspects for new and existing buildings, and requirements for a marketable circulation heat pump are developed.

For the reference building, the calculated heat loss of the hot water distribution system is 484.5 W. The potential heat extraction from the cold-water network is 329.7 W. The usable potential depends heavily on the location. Alpine regions show limited potential due to low cold-water temperatures, while urban areas with higher summer temperatures may be more suitable. The hydraulic function was qualitatively confirmed on the test rig. The heat pump extracted heat from the cold-water system and transferred it to the upper part of the hot-water storage tank. The thermal stratification remained largely stable. Due to a compressor failure, it was not possible to complete the full series of measurements.

The calculation model shows that the circulation heat pump can provide energy to the hot water storage tank. In the model, 4,380 kWh/a of heat are supplied to the storage tank, of which 2,947 kWh/a are extracted from the cold-water system. This results in an annual performance factor of 3.06. The main charging system remains necessary, supplying 11,066 kWh/a. For future implementation, hydraulically decoupled integration, stable control, SVGW-compliant drinking water components, electrical power measurements, and further reproducible measurements are required.

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1 Introduction

Climate change is increasingly leading to higher ambient temperatures, which in turn affects the drinking water supply. Especially during the summer months, rising soil and indoor temperatures can cause cold water to warm up both in the public distribution network and inside buildings. Elevated cold water temperatures can compromise user comfort and, above certain temperature thresholds, can also pose hygiene risks.

1.1 Motivation

The motivation for this work is to counteract rising drinking water temperatures and bring the cold water into a comfortable temperature range. At the same time, the energy potential present in the cold water is to be utilized to offset a portion of the heat losses from the hot water circulation. This concept allows the plumbing system to be energy-optimized and enables the use of existing heat within the building.

1.2 Research Question and Objective

The objective of this thesis is to investigate the hydraulic integration of a heat pump into a cold- and hot-water network to offset a portion of the resulting circulation heat losses. In particular, the potential of the cold-water network as a heat source is evaluated, since drinking water can become increasingly warm both inside the building and as a result of elevated ambient temperatures in the summer. By utilizing the cold-water distribution system as a heat source, heat is to be extracted from the drinking water and used to reheat the hot-water system.

In addition, the hydraulic, energy-related, and hygienic limitations of the system are examined, and possible optimization measures are identified. This raises the following questions:

- How can a circulation heat pump be integrated into a cold and hot water network in a hydraulically stable manner?
- What is the thermal potential of the cold water network as a heat source?
- What are the effects on the hot water system and storage stratification?
- What requirements must be met for a future market-ready implementation?

1.3 Approach and Structure of the “ ” Thesis

To answer these questions, theoretical principles, experimental investigations, and a simplified calculation model are combined. First, the requirements for drinking water, the fundamentals of hot water circulation, and the potential of the cold water network as a heat source are examined. Building on this, a test setup for the hydraulic integration of a circulation heat pump between the cold and hot water systems is described.

The experimental investigation includes commissioning and initial test measurements on the test setup. Due to a failure of the heat pump, it was not possible to conduct a complete series of measurements. In addition, a simplified simulation model is used to theoretically analyze the thermal energy flows in the building.

This thesis is structured into theoretical foundations, a description of the test setup, commissioning and test measurements, simulation, discussion, and conclusions and outlook.

1.4 Scope

This thesis is limited to the hydraulic integration and qualitative evaluation of a circulation heat pump in the cold and hot water systems. The focus is on using the building's internal cold water system as a heat source to partially offset circulation heat losses.

Due to the failure of the heat pump, it was not possible to conduct complete measurement series or repeat measurements. Therefore, no definitive conclusions can be drawn regarding efficiency, the annual performance factor, or long-term behavior.

Microbiological analyses and a formal SVGW compliance test are not part of this study. The hygienic assessment is based on normative requirements, temperature limits, and design considerations.

2 Theoretical al Foundations

2.1 Introduction to Theoretical al Fundamentals

This chapter provides the theoretical fundamentals necessary for the analysis of the circulation heat pump and its integration into a drinking water system. A central aspect of this chapter is the examination of the requirements for drinking water in buildings, particularly with regard to temperature, hygiene, and comfort. Furthermore, a representative multi-family residence is defined to serve as the basis for calculations of hot and cold water distribution. In addition, the hydraulic and thermal fundamentals of circulation are examined, taking into account both heat losses on the hot-water side and heat inputs into the cold-water network. Furthermore, the system’s energy balance and the usable energy from the cold-water network are analyzed. Another focus is on the examination of hot water heating systems, heat pumps, and possible alternative sources and sinks. The objective of this chapter is to establish the relevant theoretical foundations for the subsequent design, operation, and evaluation of the circulation heat pump test rig. A discussion of the theoretical foundations follows.

2.2 Methodology for Theoretical Foundations

Relevant standards, guidelines, technical literature, and previous bachelor’s theses are consulted to develop the theoretical foundations. The requirements for drinking water in buildings are examined based on hygienic, thermal, and comfort-related specifications. In particular, the permissible temperatures in the cold- and hot-water systems as well as microbiological limit values are considered. A representative multi-family residence serves as the basis for the calculations; its parameters are taken from previous studies and adapted for the present investigation. The sizing of the hot and cold water distribution systems is based on Guideline W3 and supplemented by additional calculation tools. To determine heat losses and heat gains, pipe lengths, pipe dimensions, insulation thicknesses, and ambient temperatures are analyzed. In addition, energy balances are prepared to evaluate the usable potential of the cold-water system as a heat source for the circulation heat pump. The fundamentals of hot water heating, heat pumps, and possible alternative sources and sinks are supplemented by technical literature and previous studies. Finally, the findings are synthesized to establish a theoretical basis for the subsequent evaluation of the test setup.

2.3 Results: Theoretical Principles

2.3.1 Drinking Water Requirements in the “ ” Building

This subsection describes the most important requirements for drinking water in buildings. It addresses rising cold-water temperatures, user comfort when using drinking water, as well as hygienic requirements and relevant temperature limits.

Rise in Cold Water Temperatures

The temperatures of surface water, groundwater, and—in particular—drinking water in distribution networks are rising significantly as a result of climate change. The water can warm up considerably, especially within the pipe networks, as it increasingly approaches the ground temperature with longer residence times. Long pipeline routes and stagnant sections of the network further exacerbate this effect. As a result, elevated drinking water temperatures are already occurring today and are projected to rise further to over 20 °C by 2060, thus posing a significant challenge for future drinking water supply (Degen & Kunz, 2025).

Comfort

Cold drinking water is perceived differently depending on personal preference. However, a range of approximately 8 °C to 15 °C is generally considered the ideal drinking temperature, as water within this range is perceived as pleasant and refreshing (von Euw, 2025). Temperatures above 15 °C, on the other hand, are often perceived as less refreshing or flat. Very cold water can also have unpleasant effects: at temperatures below 3 °C, it can cause what is known as “brain freeze,” while regular consumption of water below 5 °C can contribute to gastrointestinal discomfort (Degen & Kunz, 2025).

Hygiene Requirements and Water Temperatures

To ensure compliance with drinking water hygiene requirements, the temperature in the cold-water system must not exceed 25 °C (SIA, 2020). To prevent rapid proliferation of *Pseudomonas aeruginosa*, the cold-water temperature should not exceed 17 °C (Degen & Kunz, 2025).

On the hot water side, temperature requirements are specified by SIA 385/1 and the SVGW. The following temperatures apply to hot water distribution systems with heat-retaining pipes:

Table 1: Temperature Overview for Hot Water Systems with Insulated Pipes

	SIA 385/1 (2020)	W3/E3
Storage tank / FWS	**	60 °C
Hot Water Distribution System	55 °C	55 °C
Tap	50 °C	50 °C
Cold water	≤ 25 °C	≤ 25 °C

**It is noteworthy that SIA 385/1:2020 does not specify either a specific storage temperature or a temperature at the outlet of a fresh water station (FWS) for systems with heated pipes. These values must therefore be calculated by the designer (suissetec, 2020-a).

Microbiological contamination in drinking water systems is assessed using the CFU value. CFU stands for colony-forming units and describes the number of viable microorganisms in a water sample. The relevant committees of the SVGW and the TBDV have established the following limit values for this purpose (TBDV, 2016):

Table 2 Limit values for microbiological parameters. Source (TBDV, 2016):

Parameter	Maximum Values	Scope	Note
Aerobic, mesophilic microorganisms	100 CFU/mL	Sample	Incubated at 30 °C
Aerobic, mesophilic bacteria	300 CFU/mL	Distribution network	Incubated at 30 °C
Escherichia coli	0 CFU/mL	Building plumbing	
Enterococci	0 CFU/ml	Plumbing system	
Pseudomonas aeruginosa	0 CFU/mL	Public swimming pools	Bathroom in general

2.3.2 , a representative building

This thesis is based on a representative apartment building that was previously used in the bachelor’s theses BAT_G_22_11 and BAT_G_25_09. This reference building is used to study air circulation and allows for the analysis of a realistic real-world scenario. This building type was chosen because apartment buildings are widespread in Switzerland and thus have high practical relevance. For this thesis, certain parameters of the building were slightly adjusted compared to the aforementioned theses.

The apartment building has the following parameters:

- Four stories and one basement level
- Floor-to-floor height of 3 m
- Two apartments per floor, each housing two people (8 apartments, 16 people)
- The basement is unheated (18°C)
- Shaft temperature: 25 °C
- Usage per apartment:
  - Hot water: 7 LU
  - Cold water: 9 LU
- Pipe volume approx. 50 l
- Hot water storage tank capacity: 800 l
- Number of circulation loops: 2

2.3.3 Circulation

The data from the representative multi-family residence in Section 2.3.2 serves as the basis for the design and calculation of the circulation system. The pipe lengths were adopted from the reference building. For the circulation calculation, the hot and cold water systems are divided into three groups to enable the calculation of both the heat losses of the individual groups and the hydraulic balancing.

In the first step, the plumbing system is sized using the simplified method in accordance with Guideline W3. In a second step, the dimensions determined by the simplified method are optimized using the pressure loss tool from Nussbaum (R. Nussbaum AG, 2022-a). During the design process, care is taken to comply with the SVGW's maximum permissible flow velocities for outlet, floor distribution, and distribution pipes.

The pipe layout diagram shown in Appendix A.1 serves as the basis for the circulation calculations. It illustrates the schematic layout of the representative building with the hot and cold water pipes under consideration and is used to divide the network into calculation groups as well as to determine pipe lengths, heat losses, and hydraulic pressure losses.

Based on the pipe diameters and the lambda values of the various insulation materials, the required insulation thickness is determined in accordance with Table 18 of the Suissetec technical bulletin. The corresponding values have been adopted and are presented in Table 3 below.

Table 3 Minimum insulation thicknesses according to cantonal energy laws and recommended insulation thicknesses for drinking water and hot water pipes at +60 °C (Suissetec, 2023)

	Insulation Thicknesses According to Cantonal Energy Laws Temperatures from 30°C to 90°C		ISOLSUISSE recommendation for insulation thicknesses at hot water temperatures from 60°C	
Nominal pipe diameter	λ ≤ 0.03 W/mK	λ > 0.03 - ≤0.05 W/mK	λ ≤ 0.03 W/mK	λ > 0.03 - ≤0.05 W/mK
DN 10	30 mm	40 mm	30 mm	60 mm
DN 15	30 mm	40 mm	30 mm	60 mm
DN 20	40 mm	50 mm	40 mm	60 mm
DN 25	40 mm	50 mm	40 mm	60 mm
DN 32	40 mm	50 mm	40 mm	80 mm

The heat transfer coefficient (U_{tot}) of the insulated piping is then calculated using the formula for heat transfer through cylindrical pipe layers.

$$U_{tot} = \frac{2 * \pi}{\frac{1}{h_i * r_{i.R.}} + \frac{1}{\lambda_1} ln \frac{r_{a.R.}}{r_{i.R.}} + \frac{1}{\lambda_2} ln \frac{r_{tot}}{r_{a.R.}} + \frac{1}{h_a * r_{tot}}}$$

This calculation takes into account heat transfer on the inner surface, thermal conduction through the individual layers, and heat transfer to the environment. The calculation uses the pipe dimensions, the insulation thickness, the lambda value of the insulation, and the ambient conditions. The variables used are defined as follows: $r_{i,R}$ describes the inner radius of the pipe, $r_{a,R}$ the outer radius of the pipe, and r_{tot} the total radius of the pipe, including the thermal insulation. The heat transfer coefficient on the inner side is denoted by h_i, while h_a describes the heat transfer coefficient on the outer side. The thermal conductivity of the pipe material is denoted by λ_1 and that of the thermal insulation by λ_2. (von Euw, 2022)

The temperature difference, which determines the heat loss, is derived from the difference between the medium temperature and the assumed ambient temperature of the representative building. Using this temperature difference and the calculated heat output per meter, the effective heat output of the respective pipe can then be determined.

The additional parameters, such as the temperature difference and the effective heat emission, are derived from the previous calculation steps and are used to determine the total heat loss as well as heat gains from the hot and cold water circulation systems.

2.3.4 Energy Balance

In this subsection, the cold-water and hot-water distribution systems are analyzed, and the energy balances derived from the calculated heat losses and heat gains are compared.

Heat Loss in the Hot Water Distribution System

To determine the heat loss of the hot water distribution system, a circulation calculation—described in Section 2.3.3—was performed. This calculation assumes a temperature difference of 3 K between the hot water supply and the hot water return. This results in an effective heat dissipation of 484.5 W.

In addition, a flow velocity of 0.2 m/s is maintained in all sections of the hot water return line to prevent stagnation. Table 4 below shows the calculation of heat losses in the hot water distribution system.

Table 4 Circulation Calculation, Determination of Heat Dissipation in the Hot Water Distribution System

Section No.	Pipe Diameter	Section Length	Lambda value	Insulation thickness	Heat output	Medium temperature	Ambient temperature	Temperature difference	Effective heat output
	mm	m	W/(m*K)	mm	W/(m*K)	°C	°C	K	W
Group 1								**Total Group 1**	**36.03**
1.1	28	4.0	0.032	50	0.128	58	18	40	20.49
1.2	15	4.0	0.032	40	0.105	55	18	37	15.55
Group 2								**Total Group 2**	**202.31**
2.1	18	3.0	0.032	50	0.104	57	25	32	9.97
2.2	18	3.0	0.032	50	0.104	57	25	32	9.97
2.3	22	3.0	0.032	50	0.114	57	25	32	10.93
2.4	28	1.4	0.032	50	0.128	56	25	31	5.56
2.5	12	10.4	0.032	40	0.095	56	25	31	30.75
2.6	28	15	0.032	50	0.128	58	18	40	76.82
2.7	15	15	0.032	40	0.105	55	18	37	58.30
Group 3								**Total Group 3**	**246.17**
3.1	18	3.0	0.032	50	0.104	57	25	32	9.97
3.2	18	3.0	0.032	50	0.104	56	25	31	9.66
3.3	22	3.0	0.032	50	0.114	55	25	30	10.25
3.4	28	1.4	0.032	50	0.128	55	25	30	5.38
3.5	12	10.4	0.032	40	0.095	56	25	31	30.75
3.6	28	20.0	0.032	50	0.128	58	18	40	102.43
3.7	15	20.0	0.032	40	0.105	55	18	37	77.73
								Total loss	484.51 W

Heat Input from Cold Water Distribution

For the cold water distribution system, the same pipe dimensions and lengths are assumed as for the hot water distribution system. With PIR shell insulation having a thickness of 30 mm, the heat input from the environment into the cold water is 236 W. The cold water can warm up by up to 0.9 °C due to this heat absorption.

Table 5 below shows the calculation of heat gains in the cold water system.

Table 5: Calculation of Heat Gains in the Cold Water Network During Circulation Operation with PIR Shell Insulation

Calculation of Heat Gains											
Section No.	Pipe Diameter mm	Inner Diameter mm	Section Length m	Lambda value (PIR) W/(m*K)	Insulation thickness mm	Heat absorption W/(m*K)	Volume flow m³/s	Fluid temperature °C	Ambient temperature °C	Temperature difference K	Effective heat absorption W
				0.027	30		at least 0.2 m/s				
Group 1										Total Group 1	12.59
1.1	28	25.6	4.0	0.027	30	0.134	0.0001029	4.00	18	14	7.49
1.2	15	13.0	4.0	0.027	30	0.097	0.0000265	4.81	18	13.19	5.10
Group 2										Total Group 2	93.17
2.1	18	16.0	3.0	0.027	30	0.106	0.0000402	4.16	25	20.84	6.60
2.2	18	16.0	3.0	0.027	30	0.106	0.0000402	4.12	25	20.88	6.62
2.3	22	19.6	3.0	0.027	30	0.117	0.0000603	4.09	25	20.91	7.35
2.4	28	25.6	1.4	0.027	30	0.134	0.0001029	4.08	25	20.92	3.92
2.5	15	13.0	10.4	0.027	30	0.097	0.0000265	4.20	25	20.80	20.91
2.6	28	25.6	15	0.027	30	0.134	0.0001029	4.02	18	13.98	28.04
2.7	15	13.0	15	0.027	30	0.097	0.0000265	4.39	18	13.61	19.74
Group 3										Total Group 3	109.01
3.1	18	16.0	3.0	0.027	30	0.106	0.0000402	4.18	25	20.82	6.60
3.2	18	16.0	3.0	0.027	30	0.106	0.0000402	4.14	25	20.86	6.61
3.3	22	19.6	3.0	0.027	30	0.117	0.0000603	4.11	25	20.89	7.34
3.4	28	25.6	1.4	0.027	30	0.134	0.0001029	4.10	25	20.90	3.91
3.5	15	13.0	10.4	0.027	30	0.097	0.0000265	4.22	25	20.78	20.89
3.6	28	25.6	20.0	0.027	30	0.134	0.0001029	4.02	18	13.98	37.39
3.7	15	13.0	20.0	0.027	30	0.097	0.0000265	4.41	18	13.59	26.27
							Warm-up	0.81 °C		Total input	214.76 W

To enable the cold water to be used more efficiently as a source for the heat pump, the distribution network can be insulated with synthetic rubber to allow for greater heat transfer from the environment to the cold water. For this reason, the distribution system is insulated with the insulation material “ArmaFlex,” which has a lambda value of 0.036 W/m*K. The recommended insulation thickness for the synthetic rubber to prevent condensation is 13 mm at a room temperature of 25 °C and a relative humidity of 60%. This recommended insulation thickness is shown in Table 6 below.

Table 6 Recommended insulation thicknesses for preventing condensation in cold drinking water pipes (suissetec, 2023)

Insulation in mm														
Pipe diameter	17	22	28	35	42	48	60	76	89	114	140	168	175	219
PIR	30	30	30	30	30	30	30	30	30	30	30	30	30	30
Synthetic rubber	13	13	13	13	13	13	13	13	13	13	13	13	13	13

For continuous operation and under unfavorable physical conditions, an insulation thickness of 19 mm is recommended, depending on the medium temperature. Due to the low drinking water temperature of up to 4 °C, the distribution network is therefore insulated with 19 mm ArmaFlex (suissetec, 2023). The corresponding insulation thicknesses are shown in Table 7 below.

Table 7 Recommended insulation thicknesses, taking into account commercially available insulation thicknesses (suissetec, 2023)

Fluid temperature[1] : +10 °C, room temperature: +25 °C, relative humidity: 60%

Insulation in mm														
Pipe diameter	17	22	28	35	42	48	60	76	89	114	140	168	175	219
λ ≤ 0.03	30[2]	30[2]	30[2]	30[2]	30[2]	30[2]	30[2]	30[2]	30[2]	30[2]	30[2]	30[2]	30[2]	30[2]
λ > 0.03 - ≤ 0.04	19	19	19	19	19	19	19	19	25	25	32	32	32	38
λ > 0.04 - ≤ 0.05	30	30	30	30	30	30	30	40	40	40	40	40	50	60

1- The medium temperature refers to the supply line; the return line must be insulated to the same standard as the supply line.

2- Insulation made of PIR insulation materials must also be equipped with a vapor barrier of ≥50 m (μ×s).

With 19 mm of ArmaFlex insulation, a heat gain of
1.23 °C. The cold water temperatures do not fall within the microbiologically critical range of >17 °C. With this heat gain, a heat absorption of 329.7 W can occur throughout the entire chilled water system. This heat is available on the source side or on the evaporator side of the heat pump. The results mentioned are shown in Table 8 below.

Table 8: Determination of Heat Gains in the Chilled Water System During Circulation Operation with Minimal Insulation

Calculation of Heat Gains											
Section No.	Pipe Diameter	Inner Diameter	Section Length	Lambda value (synthetic rubber)	Insulation thickness	Heat absorption	Volume flow	Fluid temperature	Ambient temperature	Temperature difference	Effective heat absorption
	mm	mm	m	W/(m*K)	mm	W/(m*K)	m³/s	°C	°C	K	W
				0.036	19		at least 0.2 m/s				
Group 1										**Total Group 1**	**19.28**
1.1	28	25.6	4.0	0.036	19	0.210	0.0001029	4.00	18	14	11.76
1.2	15	13.0	4.0	0.036	19	0.147	0.0000265	5.23	18	12.77	7.52
Group 2										**Total Group 2**	**143.03**
2.1	18	16.0	3.0	0.036	19	0.162	0.0000402	4.25	25	20.75	10.10
2.2	18	16.0	3.0	0.036	19	0.162	0.0000402	4.19	25	20.81	10.13
2.3	22	19.6	3.0	0.036	19	0.182	0.0000603	4.14	25	20.86	11.37
2.4	28	25.6	1.4	0.036	19	0.210	0.0001029	4.13	25	20.87	6.14
2.5	15	13.0	10.4	0.036	19	0.147	0.0000265	4.31	25	20.69	31.68
2.6	28	25.6	15	0.036	19	0.210	0.0001029	4.03	18	13.97	44.01
2.7	15	13.0	15	0.036	19	0.147	0.0000265	4.59	18	13.41	29.61
Group 3										**Total Group 3**	**167.36**
3.1	18	16.0	3.0	0.036	19	0.162	0.0000402	4.28	25	20.72	10.08
3.2	18	16.0	3.0	0.036	19	0.162	0.0000402	4.22	25	20.78	10.11
3.3	22	19.6	3.0	0.036	19	0.182	0.0000603	4.18	25	20.82	11.35
3.4	28	25.6	1.4	0.036	19	0.210	0.0001029	4.16	25	20.84	6.13
3.5	15	13.0	10.4	0.036	19	0.147	0.0000265	4.34	25	20.66	31.63
3.6	28	25.6	20.0	0.036	19	0.210	0.0001029	4.03	18	13.97	58.69
3.7	15	13.0	20.0	0.036	19	0.147	0.0000265	4.63	18	13.37	39.38
							Warm-up	1.23 °C		Total input	329.67 W

A comparison of insulation materials shows that cold-water supply pipes insulated with synthetic rubber can absorb significantly more heat from the environment.

Table 9 Comparison of heat gains in cold-water circulation using different insulation materials

-	PIR shell insulation	ArmaFlex insulation
Lambda value	0.027 W/(m*K)	0.036 W/(m·K)
Insulation thickness	30 mm	19 mm
Heat absorption	214.76 W	329.67 W

As already described in Section 2.3.1, the ideal drinking water temperature ranges from 8 °C to 15 °C. However, drinking water temperatures below 5 °C can lead to reduced comfort and possible gastrointestinal complaints.

Since this study assumes a minimum drinking water temperature of 4 °C, such comfort limitations cannot be completely ruled out. Although the focus of this study is on energy-related aspects, the effects on the comfort range of the cold water are also evaluated in the calculation model.

After analyzing the calculations, an interim conclusion is drawn. Using a sensitivity analysis of various parameter changes, it was determined that, in particular, adjustments to the insulation and the pipe lengths of the supply network have the greatest influence on the system. Since extending the supply pipes is neither practical nor economical and incurs additional costs, this option is not considered further. By reducing the insulation thickness on the cold water side, however, additional energy can be specifically absorbed into the system.

Energy from the cold-water network

The reference building, which has eight apartments, is home to 16 people. Daily water consumption per person is 142 liters, consisting of 97 liters of cold water and 45 liters of hot water. This results in a daily cold water demand of 1,552 liters for the entire building. However, for further calculations and for creating the draw-off profile, an occupancy factor of 80% is applied, resulting in an adjusted daily cold water demand of 1,242 liters (SIA, 2024).

For the cold water draw-off profile, the temporal distribution of domestic hot water demand according to SIA 385/2 is used for simplicity. This is a model assumption, as SIA 385/2 primarily describes hot water demand (SIA, 2025). It is assumed that the percentage distribution of cold water consumption over the course of a day is similar to that of domestic hot water demand.

Therefore, the values from SIA 385/2 are also applied to cold water. The following figures show a bar chart and a cumulative line chart of domestic hot water demand in residential buildings according to SIA 385/2. Figure 1 shows water demand on weekdays from Monday through Friday, while Figure 2 shows water demand on Saturday and Sunday. Table 10 also presents the values from these figures in tabular form.

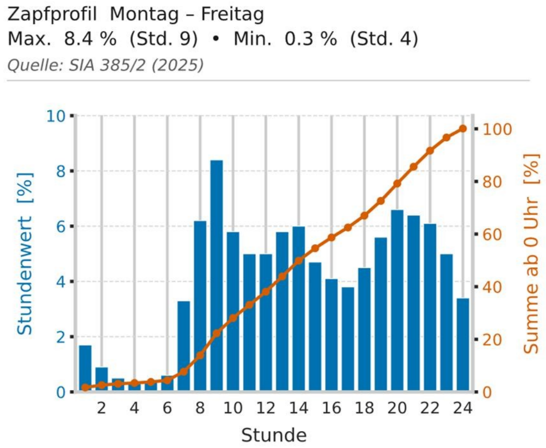

Figure 1 Bar and cumulative line chart of DHW demand in residential buildings. Monday through Friday (SIA, 2025)

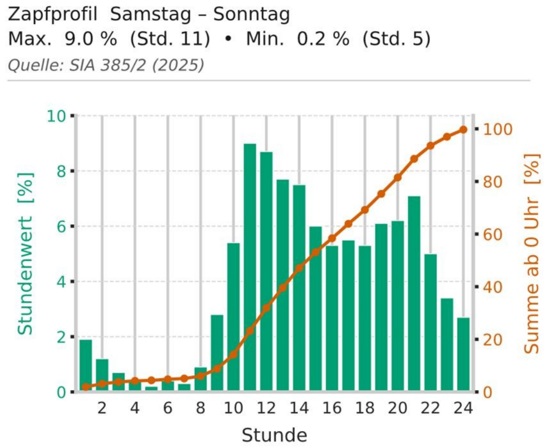

Figure 2 Bar and cumulative line chart of DHW demand in residential buildings. Saturday through Sunday (SIA, 2025)

Table 10: Static distribution of hot water withdrawal for residential buildings (SIA, 2025).

Static distribution of hot water withdrawal in residential buildings								
Time of Day	Mon–Fri				Sat–Sun			
	Hourly value [%]	Hourly value [l]	Total since midnight [%	Total since midnight [l]	Hourly value [%]	Hourly value [l]	Total since midnight [%]	Total since midnight
1	1.7%	21.1 l	1.7%	21.1 l	1.9%	23.6 l	1.9%	23.6 l
2	0.9%	11.2 l	2.6%	32.3 L	1.2%	14.9 l	3.1%	38.5 l
3	0.5%	6.2 l	3.1%	38.5 L	0.7%	8.7 l	3.8%	47.2 l
4	0.3%	3.7 l	3.4%	42.2 L	0.4%	5.0 l	4.2%	52.1 l
5	0.4%	5.0 L	3.8%	47.2 l	0.2%	2.5 l	4.4%	54.6 l
6	0.6%	7.4 l	4.4%	54.6 l	0.4%	5.0 l	4.8%	59.6 l
7	3.3%	41.0 l	7.7%	95.6 l	0.3%	3.7 l	5.1%	63.3 l
8	6.2%	77.0 l	13.9%	172.6 l	0.9%	11.2 l	6.0%	74.5 l
9	8.4%	104.3 l	22.3%	276.9 l	2.8%	34.8 l	8.8%	109.3 l
10	5.8%	72.0 l	28.1%	348.9 l	5.4%	67.0 l	14.2%	176.3 l
11	5.0%	62.1 l	33.1%	411.0 l	9.0%	111.7 l	23.2%	288.1 l
12	5.0%	62.1 l	38.1%	473.0 l	8.7%	108.0 l	31.9%	396.1 l
13	5.8%	72.0 l	43.9%	545.1 l	7.7%	95.6 l	39.6%	491.7 l
14	6.0%	74.5 l	49.9%	619.6 l	7.5%	93.1 l	47.1%	584.8 l
15	4.7%	58.4 l	54.6%	677.9 L	6.0%	74.5 l	53.1%	659.3 l
16	4.1%	50.9 L	58.7%	728.8 L	5.3%	65.8 l	58.4%	725.1 l
17	3.8%	47.2 l	62.5%	776.0 l	5.5%	68.3 l	63.9%	793.4 l
18	4.5%	55.9 l	67.0%	831.9 L	5.3%	65.8 l	69.2%	859.2 l
19	5.6%	69.5 l	72.6%	901.4 l	6.1%	75.7 l	75.3%	934.9 l
20	6.6%	81.9 L	79.2%	983.3 l	6.2%	77.0 l	81.5%	1,011.9 l
21	6.4%	79.5 L	85.6%	1062.8 l	7.1%	88.2 l	88.6%	1,100.1 l
22	6.1%	75.7 l	91.7%	1,138.5 L	5.0%	62.1 l	93.6%	1,162.1 l
23	5.0%	62.1 L	96.7%	1,200.6 L	3.4%	42.2 l	97.0%	1,204.4 L
24	3.4%	42.2 L	100%	1,242.8 L	2.7%	33.5 l	100%	1,237.9 L

To harness potential energy from the water network, data from various regions of Switzerland were collected and analyzed. The data from St. Moritz, Chur, and Zurich were compared.

Water temperatures in the St. Moritz and Chur areas show a similar annual pattern. The average annual water temperature varies mainly between 6 °C and 10 °C. Figures 3 and 4 show the drinking water temperatures at various measurement points in the drinking water networks of St. Moritz and Chur.

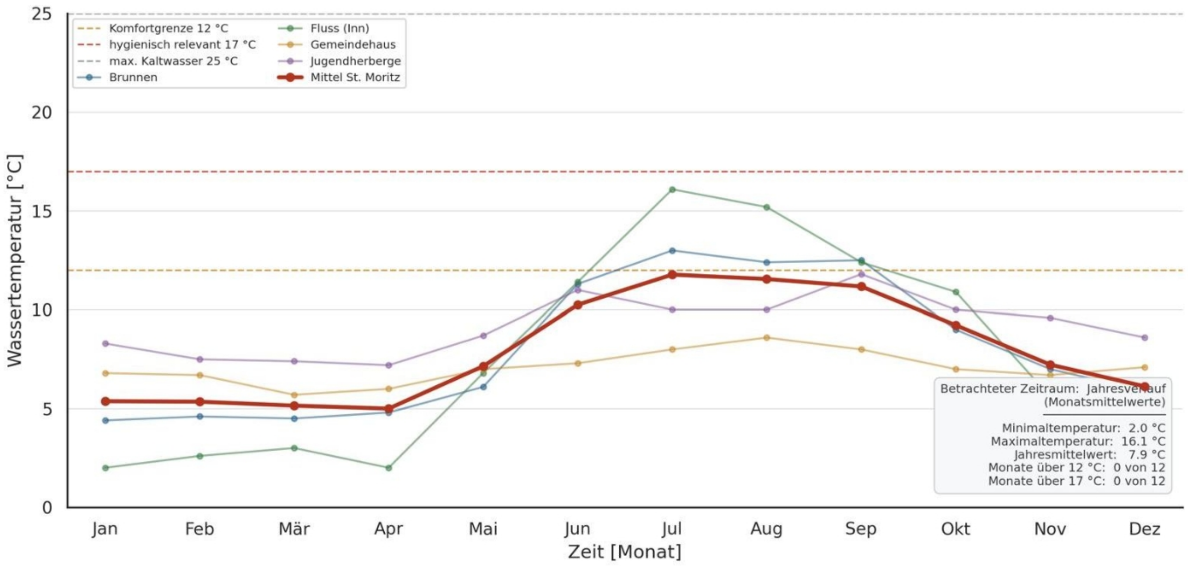

Figure 3: Line graph of annual water temperatures in St. Moritz

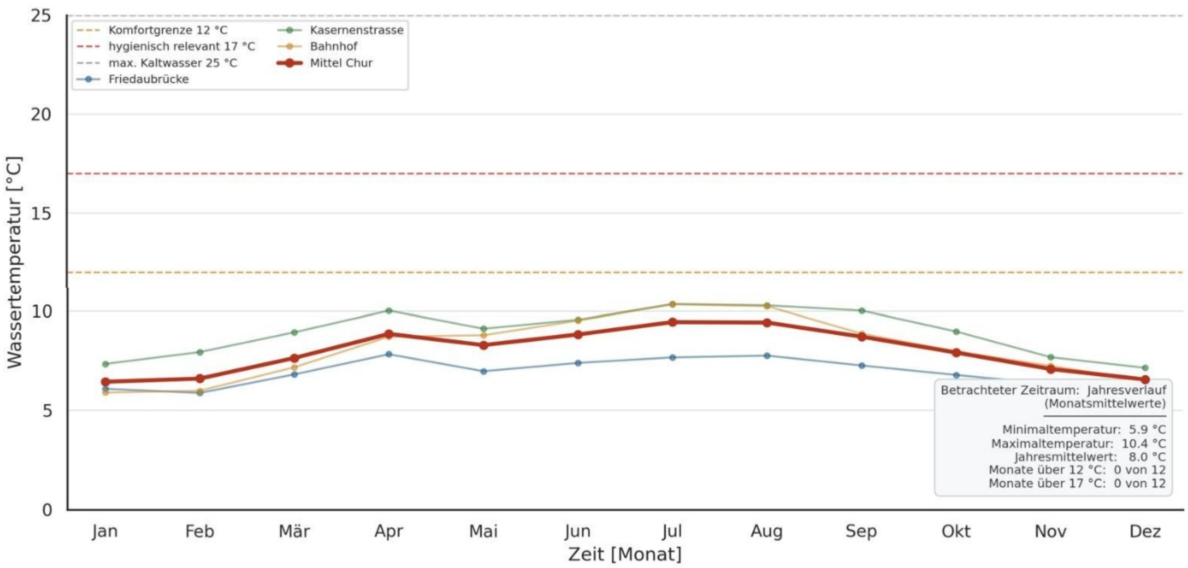

Figure 4: Line graph of annual water temperatures in Chur

The data from Zurich-Höngg differ significantly in terms of temperature levels from the data from St. Moritz and Chur. In the Höngg area, temperatures reach up to 20 °C in the summer. These are shown in Figure 5.

It should be noted that the city of Zurich uses three different water sources: groundwater, spring water, and lake water. Since all three sources are mixed together in the distribution network, varying temperature fluctuations may occur.

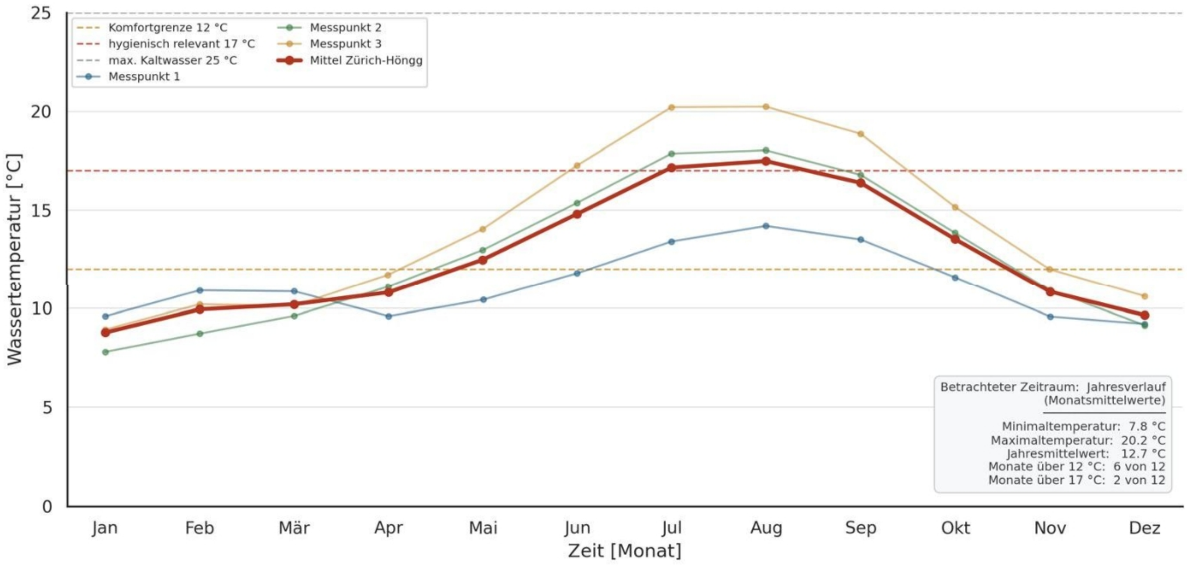

Figure 5: Line graph of annual water temperatures in Zurich-Höngg (exact measurement locations have been anonymized)

Figure 6 compares the three locations—Chur, St. Moritz, and Zurich-Höngg—in a single graph.

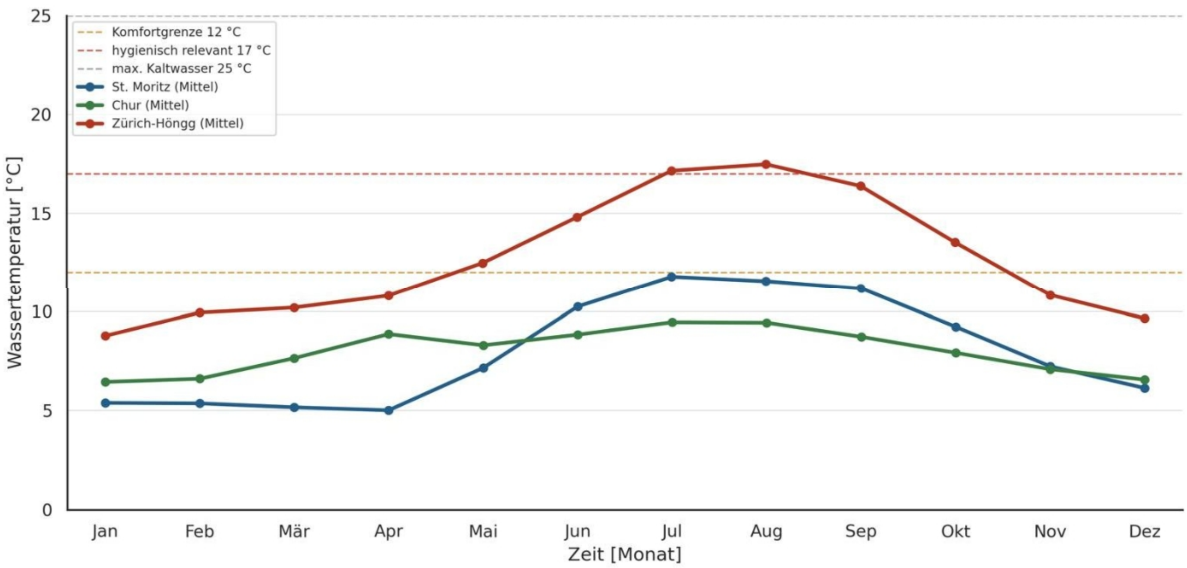

Figure 6: Line graph showing annual water temperatures at the locations Chur, St. Moritz, and Zurich-Höngg

By hydraulically separating the heat pump from the cold water circulation system using a 50-liter cold water storage tank, the volume of up to 50 liters used after each cold water withdrawal, as well as the entire piping network, can be utilized as a source for the heat pump and cooled.

An analysis of data from water utilities suggests that there is less potential for the use of circulation heat pumps in Alpine regions. In these regions, maximum cold water temperatures of around 10 °C are reached in the summer. This means that the drinking water is already within the recommended comfort range of 8 °C to 15 °C and is perceived as pleasantly cool. Figure 7 shows qualitative assessments of the potential for circulation heat pumps as well as the locations of the three water utilities under consideration.

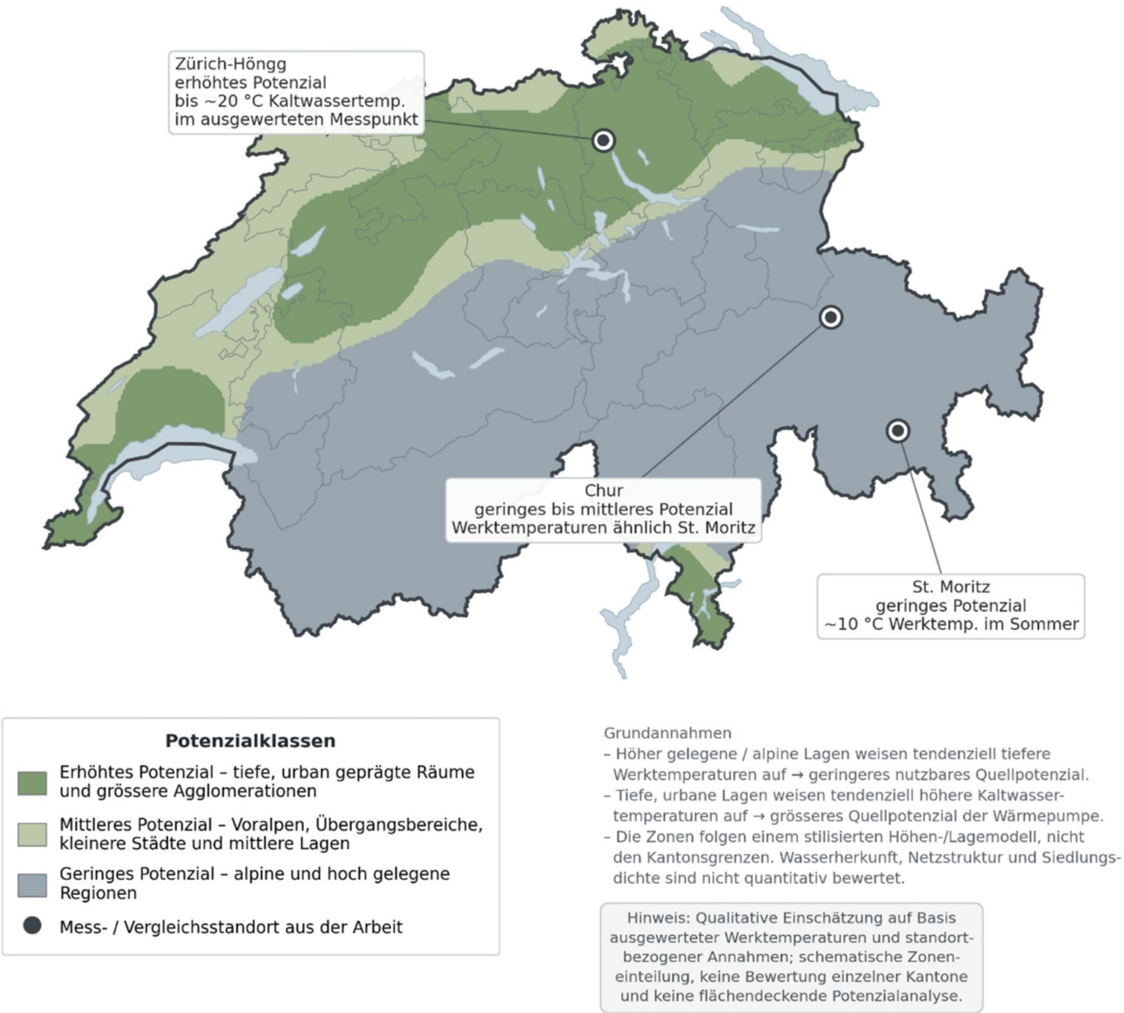

Figure 7 Qualitative potential assessment for circulation heat pumps. Data sources: swisstopo (2025) and Interactive Things (2025).

2.3.5 Hydraulics in Hot Water Circulation

Hot water circulation is a central component of larger domestic hot water systems. The purpose of circulation is to maintain the temperature within the hot water distribution network so that hot water is available at the draw-off points within a short time and hygienically critical temperature ranges are avoided. In this process, the hot water is returned to the water heater via a circulation line and reheated to the desired temperature. The SVGW W3/E3 guideline requires, in particular, the prevention of stagnation and impermissible temperature ranges in cold- and hot-water systems to ensure hygienically sound drinking water installations (SVGW, 2020).

In addition to temperature maintenance, hydraulic balancing is also crucial for the design of hot water circulation systems. The individual circulation loops exhibit different pressure losses due to varying pipe lengths, fittings, and sizing. Without balancing, an excessively high flow rate flows through hydraulically favorable loops, while hydraulically unfavorable loops may not receive sufficient flow. This can lead to insufficient return temperatures, increased heat losses, and operating conditions that are unfavorable from a hygiene perspective. Suissetec therefore describes properly performed hydraulic balancing as essential for the function, hygiene, and energy efficiency of hot water circulation systems (suissetec, 2026-a).

Balancing is achieved via circulation valves in the individual loops. A distinction can be made between hydraulically set valves and temperature-controlled valves. Hydraulically set valves limit the flow rate to a calculated setpoint and operate largely independently of temperature. Temperature-controlled circulation valves adjust the flow rate based on the water temperature in the circulation loop. At lower temperatures, the flow rate is increased; at higher temperatures, it is reduced. This allows hydraulic balancing to be combined with temperature control in the circulation network. The requirements for temperature control and hydraulic balancing are derived from the relevant standards and guidelines for domestic hot water systems (SIA, 2020; SVGW, 2020). Kemper's manufacturer documentation describes corresponding automatic circulation control valves as valves that regulate flow based on temperature (Kemper, n.d.-b).

Electronically controlled circulation valves expand this function to include sensors, actuators, and parameterization. They can measure temperature values, automatically adjust the valve position, and, in some cases, be integrated into higher-level systems. Nussbaum, for example, describes an electronically controlled circulation valve that regulates the flow rate in hot- or cold-water circulation systems based on the water temperature (R. Nussbaum AG, 2022-b). Such systems can improve the monitoring and control of circulation but do not replace the design planning, correct sizing, and commissioning of the entire system.

In addition to hygiene requirements, the circulation hydraulics have a significant impact on the energy efficiency of the entire system. Continuous circulation causes ongoing heat losses in the heated pipes. The circulation flow rate must be designed to compensate for these heat losses and maintain the required temperatures in the system. At the same time, excessively high flow rates lead to increased pump power and can further increase heat losses. If there are multiple circulation loops, hydraulic balancing is also required (SIA, 2020; SVGW, 2020; suissetec, 2026-a).

Furthermore, the return temperature of the circulation system influences storage tank management. If the circulation return is connected to the storage tank in an unfavorable manner or flows at excessively high flow rates, thermal stratification within the storage tank can be impaired. This reduces the usable temperature spread and can reduce the efficiency of water heating. This relationship is essential for the present study, as the circulation heat pump is not integrated directly into the circulation line but operates via the hot water storage tank. The aim is to minimize the impact on the existing circulation hydraulics while simultaneously providing energy support to the upper zone of the storage tank.

In this study, the utilization of circulation heat losses is of central importance. The goal is not to alter the existing circulation design, but rather to provide energy support to the hot water storage tank via a hydraulically decoupled circulation heat pump. The correct design and adjustment of the hot water circulation remain a prerequisite for hygienic and hydraulically stable operation of the entire system.

2.3.6 Hot Water Heating Systems

Domestic hot water heating must ensure comfort, hygiene, operational safety, and energy efficiency. In Switzerland, the regulatory requirements for domestic hot water systems are described in particular by SIA 385/1 and the SVGW guidelines. Relevant factors include, among others, temperature maintenance, the prevention of stagnation, and hygienically safe operation of the system (SIA, 2020; SVGW, 2020).

In larger buildings, hot water systems are often set up centrally. The domestic hot water is provided by a water heater or storage tank and distributed to the draw-off points via a distribution network. The storage tank serves to cover peak loads and ensure a reliable supply. At the same time, heat losses occur in the storage, distribution, and circulation systems, which must be compensated for by the heat generator. The planning and design of the hot water system therefore have a significant impact on energy demand (SIA, 2020; SIA, 2025).

Thermal stratification within the storage tank is important for the efficiency of central hot water systems. The upper zone of the storage tank provides the usable temperature range, while lower temperatures remain in the lower zone. Unfavorable inflow conditions, high flow rates, or recirculation can disrupt this stratification and reduce the usable temperature spread. This relationship was investigated in the bachelor’s thesis BAT_G_24_17 using a hot water test rig (Märki & Schmid, 2024).

Hot water circulation serves to maintain the temperature in the heated pipes but causes continuous heat loss. A hydraulically balanced circulation system is therefore necessary for both hygiene and energy efficiency. Without proper balancing, individual loops may be undersupplied, while others may have excessively high flow rates (suissetec, 2026-a).

Circulation losses are particularly relevant to this study. The circulation heat pump is designed to utilize existing heat from the cold-water network to support the hot-water storage tank. In doing so, the existing hot-water circulation system should be affected as little as possible from a hydraulic standpoint.

2.3.7 Heat Pumps

Heat pumps are thermodynamic systems that absorb heat at a low temperature and, using drive energy, raise it to a higher, usable temperature. This is achieved through a cycle that essentially consists of an evaporator, a compressor, a condenser, and an expansion device (Dott et al., 2018).

The efficiency of a heat pump is typically described by the Coefficient of Performance (COP). This represents the ratio of heat output to electrical power input. Efficiency depends largely on the temperature difference between the heat source and the heat sink. The smaller this temperature difference, the higher the COP can be (Dott et al., 2018).

When it comes to domestic hot water heating, heat pumps face higher demands than in low-temperature heating systems. This is due to the required hot water temperatures as well as storage and circulation losses, which must be regularly compensated for. As a result, the storage temperature, the source temperature, the charging authorization, and the hydraulic integration in particular influence the operation of the heat pump. These factors include enabling or disabling the heat pump, storage tank management, the operation of circulation pumps, maintaining minimum flow rates, and limiting cold and hot water temperatures.

Natural refrigerants are becoming increasingly important for future heat pump systems. In Switzerland, refrigerants are regulated under Annex 2.10 of the Chemicals Risk Reduction Ordinance. The primary goal of this regulation is to reduce emissions of ozone-depleting and highly global-warming-potential refrigerants (BAFU, 2025). As a result, the importance of natural refrigerants such as propane (R290) for heat pump applications is increasing.

Propane has good thermodynamic properties and a very low global warming potential. At the same time, R290 is classified as a flammable refrigerant. Safety requirements therefore depend heavily on the refrigerant charge, the installation location, accessibility, and ventilation and protective measures. EnergieSchweiz points out that, depending on the installation and charge quantity, additional safety measures may be required for propane heat pumps (EnergieSchweiz, n.d.).

Many manufacturers are therefore developing compact R290 heat pumps with reduced refrigerant charge levels to meet safety requirements. For heat pumps and refrigeration systems using low-toxicity, flammable refrigerants, suissetec states that systems using propane or isobutane with a charge of up to 1.5 kg are not subject to the requirements of EKAS Guideline 6517 "Liquefied Gas" (suissetec, 2026-b). Low refrigerant charges can thus simplify installation. At the same time, refrigeration-related challenges arise, as evaporation, condensation, superheating, and stable operation must function reliably even with a small amount of refrigerant.

For a circulation heat pump, in addition to the refrigerant, the hydraulic integration is particularly crucial. The bachelor's thesis BAT_G_25_09 showed that low-capacity heat pump systems can be sensitive to unstable flow rates, unfavorable temperature differentials, and undefined operating conditions. The heat pump must therefore be integrated in such a way that stable flow rates and controlled temperature ranges are achieved on both the source and sink sides.

In this thesis, the heat pump is not considered a primary heat generator, but rather a supplementary system for utilizing heat from the chilled water network. The goal is to use existing heat within the building system to partially offset the circulation heat losses of the hot water system. At the same time, heat is to be extracted from the chilled water. The heat pump thus forms the energy link between the chilled water network and the hot water storage tank.

2.3.8 Possible Alternative Sources and Sinks

The temperatures of the heat source and the heat sink are critical for the operation of a circulation heat pump. The efficiency of a heat pump depends largely on the temperature difference between the source and the sink. The smaller this temperature difference, the higher the theoretical COP (Dott et al., 2018). Since both the cold water temperature and the storage tank temperature fluctuate over the course of the year and during operation, different source, sink, and control conditions must be considered.

Source

Temperature Level of the Heat Source

The building's internal chilled water network serves as the primary heat source. The heat present in this network can be extracted via the heat pump and used to reheat the hot water storage tank. This cools the chilled water and simultaneously covers a portion of the circulation heat losses in the hot water system. Central hot water systems exhibit significant heat losses, particularly due to insulated pipes and circulation systems, which must be compensated for by the heat generator (SIA, 2020; SIA, 2025; suissetec, 2026-a).

The usefulness of the cold-water network as a heat source depends heavily on the cold-water temperature. The higher the source temperature, the smaller the temperature difference between the cold-water network and the hot-water storage tank. This improves the theoretical COP (Dott et al., 2018). Especially in

the summer months or in urban areas with higher ambient temperatures, the chilled water network can therefore offer significant energy potential.

At low cold-water temperatures, however, the usable potential decreases. On the one hand, less heat is available on the source side; on the other hand, the temperature difference to the hot-water side increases. This reduces efficiency (Dott et al., 2018). In addition, care must be taken to prevent the cold water from cooling to a temperature range that is unfavorable for comfort or operational purposes as a result of heat extraction. The SVGW Guideline W3/E3 describes measures to ensure drinking water hygiene in cold- and hot-water systems during construction and operation (SVGW, 2020).

Alternative Heat Sources

If the cold-water network is insufficient as a source or is already too cold, alternative heat sources can be considered. Possible additional sources include, in particular, mechanical rooms, waste heat from building services systems, or other internal low-temperature sources. These include, for example, waste heat from refrigeration systems, ventilation systems, or server rooms. The use of waste heat and low-temperature sources is beneficial for heat pump systems, as it can reduce the temperature difference and improve efficiency (Dott et al., 2018).

Such sources are highly dependent on the specific building and its use. For concrete implementation, it is therefore necessary to assess whether the source is regularly available, what the temperature level is, and how it can be integrated into the hydraulic or air-side systems. Alternative sources should therefore primarily be considered as expansion options and evaluated on a project-by-project basis.

Sink

Temperature Requirements for the Heat Sink

On the sink side, the hot water storage tank serves as the primary heat sink. The heat pump draws water from the storage tank, heats it, and returns it to the upper storage zone. This supports the upper storage zone and compensates for a portion of the circulation heat losses. Thermal stratification and the location of heat injection influence the usable storage capacity and the operational behavior of the hot water system (Märki & Schmid, 2024).

The discharge temperature is determined by the requirements for potable hot water, hygiene, and supply reliability. For potable hot water systems, SIA 385/1 and the SVGW Guideline W3/E3 must be taken into account in particular. SIA 385/1 contains basic principles and requirements for hot water systems in buildings, while the SVGW W3/E3 describes measures to ensure drinking water hygiene (SIA, 2020; SVGW, 2020). Among other things, the revised SIA 385/1 specifies requirements for temperatures, hydraulic circuits, storage tanks, piping, and water heating with heat pumps (suissetec, 2020-a).

For this project, heat transfer to the hot-water storage tank is preferred over direct integration into the circulation return line. This allows the heat pump to operate hydraulically decoupled from the hot water circulation. The flow rate and temperature spread in the heat pump circuit can be adjusted independently of the existing circulation hydraulics. The existing hot water circulation can still be sized and regulated separately. According to suissetec, a properly performed hydraulic balancing is crucial for the function, hygiene, and energy efficiency of hot water circulation systems (suissetec, 2026-a). The technical bulletin also highlights typical sources of error, circulation valves, circulation pumps, and measures to reduce heat loss (suissetec, 2026-a).

Sink Overpressure

In operating conditions with elevated cold water temperatures, targeted sink overloading or storage tank overloading can be considered. In this process, the hot water storage tank is temporarily charged beyond the normal setpoint. This creates additional storage capacity as a heat sink, which can extend the heat pump’s runtime and reduce frequent start-stop cycles. SIA 385/1 and SVGW W3/E3 specify the hygienic framework for temperature maintenance and intended operation (SIA, 2020; SVGW, 2020).

At the same time, as the storage tank temperature increases, so does the heat pump’s temperature lift. This reduces the theoretical COP (Dott et al., 2018). Increasing the storage temperature therefore represents a trade-off between longer runtime, better utilization of the cold water potential, and lower efficiency. It is only advisable if it is hygienically permissible, controllable via the control system, and compatible with the building’s storage management. Furthermore, higher storage temperatures must not result in unnecessary storage and distribution losses.

If the hot water storage tank is fully charged and cannot absorb any additional heat, the heat pump must be shut down or switched to a different operating mode. The utility room, for example, can be considered an alternative heat sink. In this case, excess heat would be released into the room air. However, this mode of operation is only sensible if it continues to provide a benefit—for example, if cooling the cold water is necessary even though the storage tank is fully charged. Otherwise, electrical energy would be consumed without providing any direct benefit to the hot water system.

Control

In this paper, the control of a circulation heat pump refers not primarily to the internal compressor control, but to the higher-level system control. This control determines whether the heat pump is enabled, disabled, or switched to a different operating mode. For domestic hot water systems, temperature maintenance, hygiene, hydraulic function, and intended operation are key constraints (SIA, 2020; SVGW, 2020; suissetec, 2026-a).

The most important control variables are the chilled water temperature, the storage tank temperature, the available storage capacity, and the minimum flow rates on the source and sink sides. In addition, the main charge of the storage tank, possible alternative sources and sinks, as well as hygienic temperature limits must be taken into account. Table 11 shows a simplified overview of possible operating conditions and control responses.

Table 11 Control States of a Circulation Heat Pump

Operating Status	Source	Sink	Control response
Storage tank below setpoint	Cold water available	Storage tank can absorb heat	Enable heat pump
Storage tank at setpoint	Cold water level increased	Storage tank overcharge permitted	Check for under- or overcharge
Storage tank fully charged	Cold water increased	Storage tank cannot absorb heat	Check for or block an alternative sink
Cold water is already cool	Source unavailable	Storage tank can absorb heat	Shut off the heat pump ; main charging takes over
Cold water too cold	Alternative source available	Storage tank can absorb heat	Check alternative source
Temperature or volume flow limit exceeded	Source or sink not permitted	Operation is not safe	Safety or hygiene lockout

Operation with a constant storage tank setpoint

In operation with a constant storage tank setpoint, the heat pump is activated when the hot-water storage tank falls below a defined setpoint and the cold-water supply is available as a source. The storage tank is then reheated to the target temperature. This strategy is simple, robust, and easy to understand. However, it utilizes the cold-water potential only when there is a simultaneous heat demand in the storage tank.

Operation with Temporary Sink Overload

In operation with temporary sink enhancement, the storage tank is temporarily charged beyond the normal setpoint when cold water potential is high. This creates additional storage capacity as a heat sink. The advantage lies in longer heat pump runtime, better utilization of high cold water temperatures, and additional cooling of the cold water.

The disadvantage is the higher temperature difference. As the storage tank temperature rises, the theoretical COP decreases (Dott et al., 2018). In addition, storage and distribution losses may increase. Exceeding the sink temperature is therefore only advisable under operating conditions where the benefits of cooling the cold water and the additional storage charge justify the loss of efficiency. At the same time, requirements for temperature control and hygiene must be met (SIA, 2020; SVGW, 2020).

Hot Water- and Cold Water-Oriented Operation Activation

The heat pump can be activated either via the hot-water storage tank or via the cold-water temperature. With hot-water-oriented activation, the charge level of the hot-water storage tank is decisive. The heat pump operates when the upper storage zone requires heat

. This strategy supports the hot water supply and compensates for circulation losses.

In cold-water-oriented activation, the cold-water temperature is the primary factor. The heat pump is operated preferentially when the cold-water temperature exceeds a defined threshold and a suitable heat sink is available. This strategy improves the efficiency of cold-water cooling but requires that the hot-water storage tank or an alternative heat sink be able to absorb heat.

For future systems, a combination of both approaches makes sense. The heat pump is operated only when usable heat is available on the source side and heat can be absorbed on the sink side. The hydraulic integration and temperature control are crucial to ensure that drinking water hygiene, operational safety, and energy efficiency are not compromised (SIA, 2020; SVGW, 2020; suissetec, 2026-a).

It follows that the cold-water network can be used as a primary heat source, but its operation depends heavily on temperature conditions. High cold-water temperatures improve the operating point and simultaneously increase the benefits of cold-water cooling. Low cold-water temperatures or a fully charged hot-water storage tank limit operation. For practical implementation, therefore, alternative sources and sinks as well as an overarching control strategy are required.

2.3.9 Summary of Previous Work in the Experimental Laboratory

In the separate module "Experimental Work," a modular and mobile test setup for cold-water circulation was designed, constructed, and documented. The goal of this work was to create a realistic simulation of a cold-water network in the laboratory that could be operated independently as well as combined with existing laboratory setups. The focus was on hydraulic integration, the selection of suitable components, and the structured documentation of the experimental setup. Measurements and energy analyses were not yet part of this work but were intentionally reserved for further investigations (Eichelberger & Paganini, 2025).

The thesis first established the theoretical foundations for circulation heat pumps, various floor-by-floor distribution systems, and Venturi flow dividers. Building on this, the cold-water network was dimensioned, taking into account pressure losses, load values, and the required minimum flow velocities. In addition, it was determined that external heat influences on the chilled water could be simulated using a heating element (Eichelberger & Paganini, 2025).

The experimental setup was constructed on a mobile table and divided into two functional areas. On one side is the distribution system with the riser zones and floor distribution points, while on the opposite side is the central unit with grid connection, storage tank, grid gains, and connection options for external devices. Using shut-off valves, screw connections, and interchangeable components, the setup was designed to allow for future adjustments, expansions, or variant tests without requiring fundamental modifications (Eichelberger & Paganini, 2025).

The design, construction, and description of the individual components were already carried out and documented in the experimental laboratory study “Hydraulic Integration of Cold-Water Circulation Loops for Heat Recovery Using a Heat Pump.” In the present study, this existing experimental setup can be utilized to carry out further adjustments, measurements, and evaluations based on it. As already noted in the previous study, the leak test, the installation of the measurement equipment, and further measurement series were still pending and planned for subsequent work (Eichelberger & Paganini, 2025).

2.4 Discussion of the Theoretical Foundations

The theoretical foundations show that the circulation heat pump has energy potential for utilizing heat from the cold-water network. Calculations for a representative multi-family residence indicate a heat loss of 484.5 W from the hot-water distribution system, while heat absorption of 329.7 W is possible via the cold-water network.

Analysis of water temperatures shows that this potential is highly dependent on location. In alpine regions such as St. Moritz and Chur, cold water temperatures remain around only 10 °C even in summer, whereas in Zurich-Höngg, temperatures can reach up to 20 °C.

The research further shows that hydraulic integration and storage stratification are crucial for stable operation. In addition, hygienic requirements and the comfort range of the cold water must be taken into account. In summary, the theoretical analysis provides a basis for investigating hydraulic feasibility, energy performance, and operational behavior on the test bench.

3 Circulating Heat Pump Test Bench

3.1 Introduction: Test Bench for a Circulation Heat Pump

The test rig used in this bachelor's thesis is designed to investigate the hydraulic integration of a circulation heat pump between the chilled water and hot water systems. The chilled water system is used as the heat source. The absorbed heat is raised to a higher temperature level via the heat pump and fed into the hot water storage tank.

The objective of the test setup is to assess the hydraulic feasibility and the thermal system behavior. The focus is on the integration of the heat pump, heat extraction from the cold water system, heat transfer to the hot water storage tank, and the effect on stratification within the storage tank.

The test setup combines the existing hot water system from the previous bachelor's thesis with the cold water network constructed in the "Experimental Work" module. This allows the heat source, heat pump, and hot water storage tank to be investigated within a single hydraulic system.

The planned measurements are based on defined operating conditions as specified in the measurement protocol. Different storage tank temperatures as well as hot and cold water draws are considered. Since the heat pump failed during the investigation, the planned measurement series could not be completed in full. The findings from the commissioning and test measurements are described in Chapter 4.

3.2 Test Bench Methodology: Circulation Heat Pump

The investigations on the test bench are based on a structured experimental methodology with defined operating conditions. The goal is a step-by-step analysis of system behavior, ranging from steady-state reference conditions to dynamic load cases involving hot and cold water withdrawal.

The test plan provides for two temperature blocks with storage tank temperatures of 50 °C and 60 °C. Within these blocks, reference conditions without heat pump operation as well as operating conditions with an active circulation heat pump are investigated. In addition, hot water draw-off, cold water draw-off, and combined draw-off conditions are included. In this context, thermal boundary conditions refer specifically to the cold water temperature, the hot water storage tank temperature, the heat pump's temperature lift, and the storage tank's charge level.

Constant flow rates are used for the draw conditions. This simplification was chosen to create reproducible laboratory conditions and to enable comparison of the individual operating conditions with one another. Short-term draw peaks are therefore not fully represented, but they are not the focus of this initial system analysis.

Data acquisition is performed using LabVIEW and LabIGE. In particular, temperatures, flow rates, and storage tank states are recorded. Additionally, steady-state criteria are defined to identify stable operating conditions. The detailed test matrix with the individual operating conditions, draw-off profiles, and temperature blocks can be found in the measurement manual in Appendix B.1.

3.3 Experimental Setup

3.3.1 : Possible Test Types

Various operating conditions are defined for the test rig to investigate the interaction between the chilled water system, the heat pump, and the hot water storage tank. The tests are intended to demonstrate how the heat pump integrates into the overall system in terms of hydraulics and temperature. The temperature-related conditions refer in particular to the chilled water temperature, the storage tank temperature, the heat pump’s supply and return temperatures, and the temperature distribution within the hot water storage tank.

The planned test series will include reference conditions without heat pump operation as well as operating conditions with an active circulation heat pump. In addition, various draw-off conditions are planned to examine the system’s behavior under defined load situations. The focus will be on the following points:

- System behavior without heat pump operation
- Heat Extraction from the Chilled Water System
- Heat transfer to the hot water storage tank
- Effect of hot water draws on storage stability
- Effect of cold water draws on the source temperature
- Behavior during simultaneous hot and cold water draw-offs

The experimental design calls for tests at storage tank temperatures of 50 °C and 60 °C. The detailed test matrix, including the individual operating conditions, draw-off profiles, and temperature blocks, can be found in the measurement protocol in Appendix B.1

3.3.2 Overview of the “ ” test rig

The test setup consists of three main areas: the cold water system, the circulation heat pump, and the hot water system with a transparent storage tank. The cold water system forms the source side of the heat pump. The hot water system with storage tank forms the sink side. There is no direct hydraulic connection between the two water systems. The energy coupling occurs exclusively via the heat pump’s refrigeration circuit.

For clarity, the test setup is first illustrated in a simplified functional diagram. Figure 8 shows the basic direction of energy flow between the chilled water system, the heat pump, and the hot water storage tank. The diagram provides an overview of the basic principle: Heat is extracted from the heat is extracted from the chilled water system, raised to a higher temperature level via the heat pump, and then fed into the hot water storage tank.

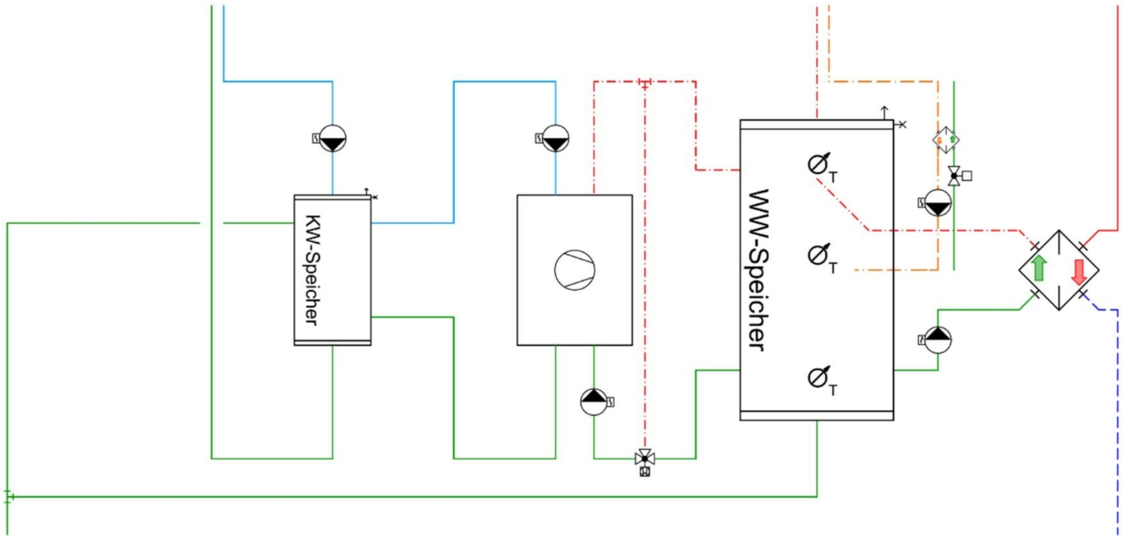

Figure 8 Simplified functional diagram of the circulation heat pump

Figure 9 shows the test setup as a more detailed hydraulic test diagram. It depicts the main supply and return connections, the hydraulic separation of the source and sink sides, as well as the relevant valves, measurement points, and components. This diagram forms the basis for understanding the measurement points, the flow rates, and the subsequent evaluation of operational behavior.

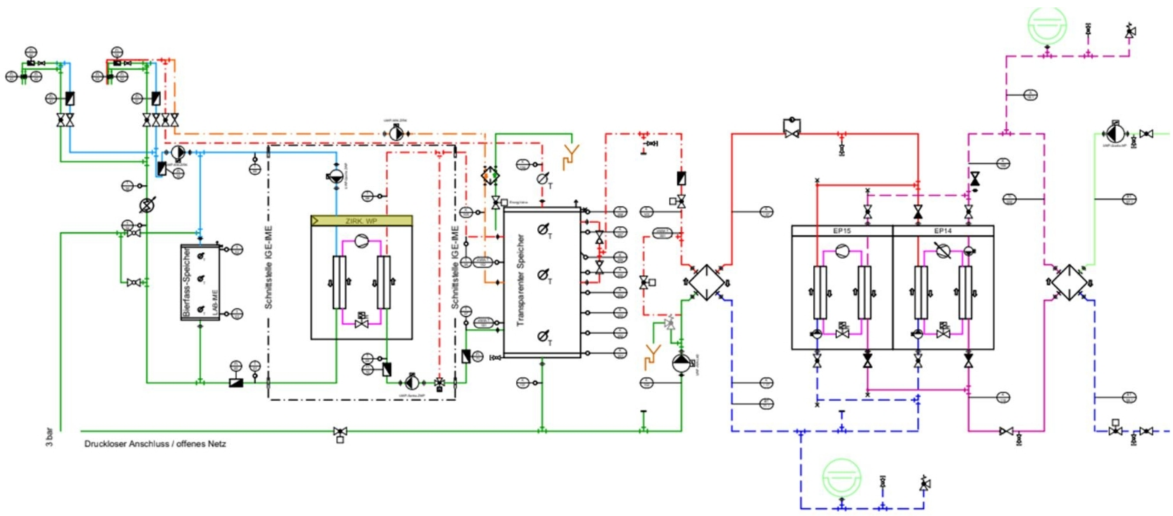

Figure 9 Hydraulic Test Diagram of the Circulation Heat Pump

Water is drawn from the lower section of the hot water storage tank, heated by the heat pump, and then returned to the upper section of the tank. This is intended to maintain the temperature of the upper storage zone specifically and to relieve the main load on the storage tank.

The hot water section is largely based on the design described in the bachelor's thesis BAT_G_24_17. Additionally, a return flow temperature maintenance system was integrated on the load side of the heat pump. This is intended to prevent excessively low return flow temperatures or unfavorable inflow conditions from disrupting the stratification within the storage tank.

3.3.3 Cold Water

The cold water system serves as the heat source for the circulation heat pump. The goal is to utilize the thermal energy present in the cold water network to partially offset the circulation losses of the hot water system. At the same time, heat is extracted from the cold water system, thereby reducing its temperature.

To simulate defined operating conditions, automatic cold-water withdrawals are implemented using solenoid valves. A constant flow rate of 420 l/h is specified for the cold-water side. This value corresponds to the withdrawal condition defined in the measurement protocol and serves to reproducibly simulate a load case in the laboratory.

The chilled water system is hydraulically separated from the hot water system. The connection between the two systems is established exclusively via the heat pump. This allows the source side to be examined independently of the hot water system. Of particular relevance are the chilled water temperature, the flow rate, and the available heat output on the evaporator side of the heat pump.

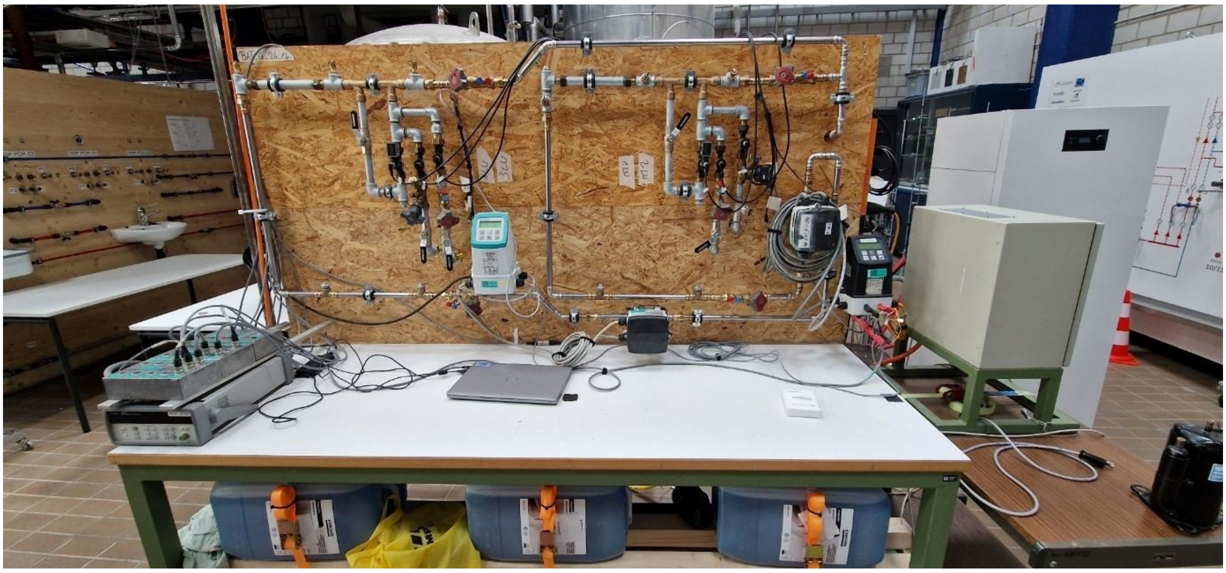

Figure 10: Chilled water system of the test setup

3.3.4 Heat Pump

A laboratory heat pump developed by the Institute of Mechanical Engineering and using the natural refrigerant propane (R290) was used for heat transfer. The heat pump was specifically designed for laboratory operation and is rated for temperature ranges that may occur in plumbing distribution systems. In the previous bachelor’s thesis BAT_G_25_09, the same basic setup was used to investigate a circulation heat pump with cold water as the heat source (De-gen & Kunz, 2025).

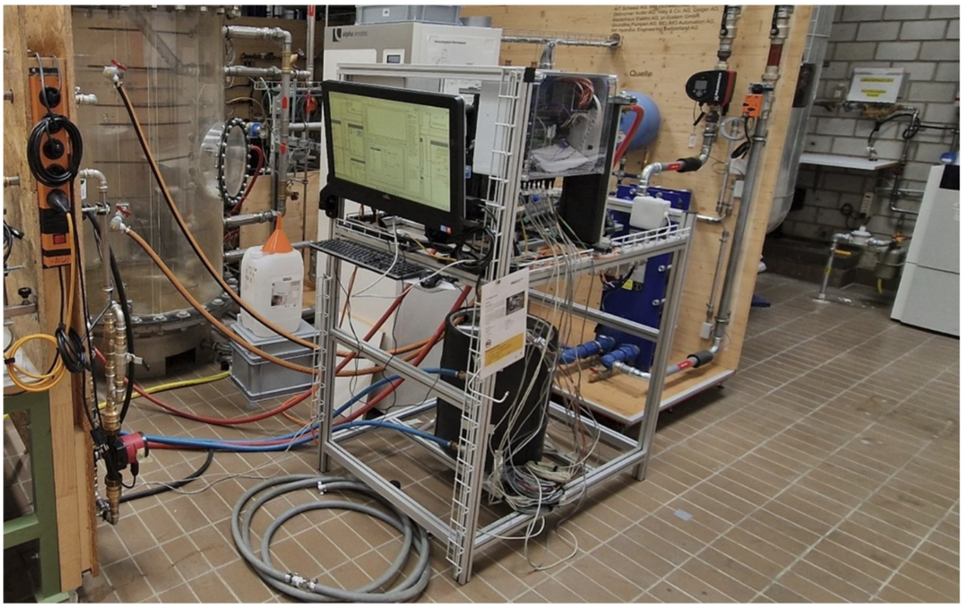

Figure 11 Circulation heat pump of the experimental setup

The laboratory heat pump has a cooling capacity of approximately 600 W. This value refers to the source side of the heat pump. For the computational model used in this thesis, the maximum heat transfer to the hot water storage tank is defined separately. The model capacity is therefore not directly equivalent to the cooling capacity of the laboratory unit.

The heat pump is hydraulically integrated between the cold water and hot water systems. Heat transfer occurs via water-filled heat exchangers on both the source and sink sides. On the evaporator side, heat is extracted from the chilled water system. On the condenser side, this heat is transferred to the hot water system together with the electrical power of the compressor. The chilled water and hot water systems remain hydraulically separated; the energy coupling occurs exclusively via the heat pump’s refrigeration circuit. BAT_G_25_09 describes that a separate circulation circuit with an unregulated circulation pump was available on both the evaporator and condenser sides. The flow rates were adjusted manually in the test setup at that time (Degen & Kunz, 2025).

The known technical specifications of the laboratory heat pump used are summarized in Table 12. Data that is not fully documented is deliberately not used as fixed parameters.

Table 12 Technical Specifications of the Laboratory Heat Pump Used

Parameter	Specification	Remarks
Refrigerant	R290 / Propane	Natural refrigerant
Cooling capacity	approx. 600 W	Specification based on the source side
Heat source	Chilled water system	Evaporator side
Heat sink	Hot water storage tank / Hot water system	Condenser side
Heat Transfer	Water-bearing heat exchangers	KW and WW hydraulically separated
Hydraulics on the source and sink sides	Each with a separate circulation circuit	in accordance with BAT_G_25_09
Circulation pumps	Grundfos Comfort PM	Unregulated, in accordance with BAT_G_25_09
Flow rate control	manually adjusted	in accordance with BAT_G_25_09
COP in measurement mode	approx. 2.5 to < 1.5	depending on operating conditions and source temperature
Condenser capacity at steady state	approx. 974 W	at approx. 263 W evaporator capacity and a COP of 1.37
Refrigerant charge	Not documented	No quantitative assessment
COP W5/W65	Not documented	No manufacturer's performance curve available

The measurements from BAT_G_25_09 show that the COP is highly dependent on operating conditions. During an initial startup cycle, an average COP of approximately 2.5 was achieved. As the chilled water temperature decreased, the COP dropped and fell below 1.5 toward the end of the measurement. At the same time, the evaporator capacity decreased. In a described steady-state condition, the evaporator capacity was still around 263 W; with a COP of 1.37, this resulted in a condenser capacity of around 974 W (Degen & Kunz, 2025).

Therefore, the source temperature, the sink temperature, the flow rates, and the supply and return temperatures on both the source and sink sides are particularly relevant for evaluating the test setup. These parameters determine whether the heat pump can operate under stable conditions and whether heat is reliably transferred from the cold water system to the hot water storage tank.

During commissioning, the hydraulic function of the system was confirmed. However, during subsequent operation, a failure of the heat pump—specifically the compressor—occurred. As a result, it was not possible to carry out complete measurement series. The exact cause could not be conclusively identified within the available project timeframe.

3.3.5 Hot Water

The hot water system consists of a transparent storage tank with multiple temperature sensors. This allows the thermal stratification within the tank to be observed during operation. The goal is to assess whether the heat input from the circulation heat pump stabilizes the upper storage zone or whether it leads to undesirable mixing.

The heat pump draws water from the lower storage zone, heats it, and then returns it to the upper storage zone. The heat is thus specifically introduced into the upper storage zone. This is intended to stabilize the top layer and reduce the necessary main charge of the storage tank.

To simulate defined load cases, hot water withdrawals with a constant flow rate are specified. A flow rate of 210 l/h is specified for the hot water side. As on the cold water side, this constant draw condition ensures the reproducibility and comparability of the laboratory tests. A dynamic daily profile is not simulated in the test setup. The chronological sequence of the planned draw conditions is documented in the measurement protocol in Appendix B.1.

During commissioning, it became apparent that some of the temperature sensors in the transparent storage tank were incorrectly labeled. The sensor positions were therefore checked and reassigned. This correction is relevant for the evaluation of storage stratification, as the position of the temperature sensors directly influences the interpretation of the storage tank conditions.

The hot water side also allows for the setting of defined circulation losses as well as the adjustment of flow rates. During commissioning, the desired flow rate could not be achieved in some cases. Contaminants in the circulation loop were identified as the cause, particularly in the area of the plate heat exchanger. This limited the system’s potential operating range.

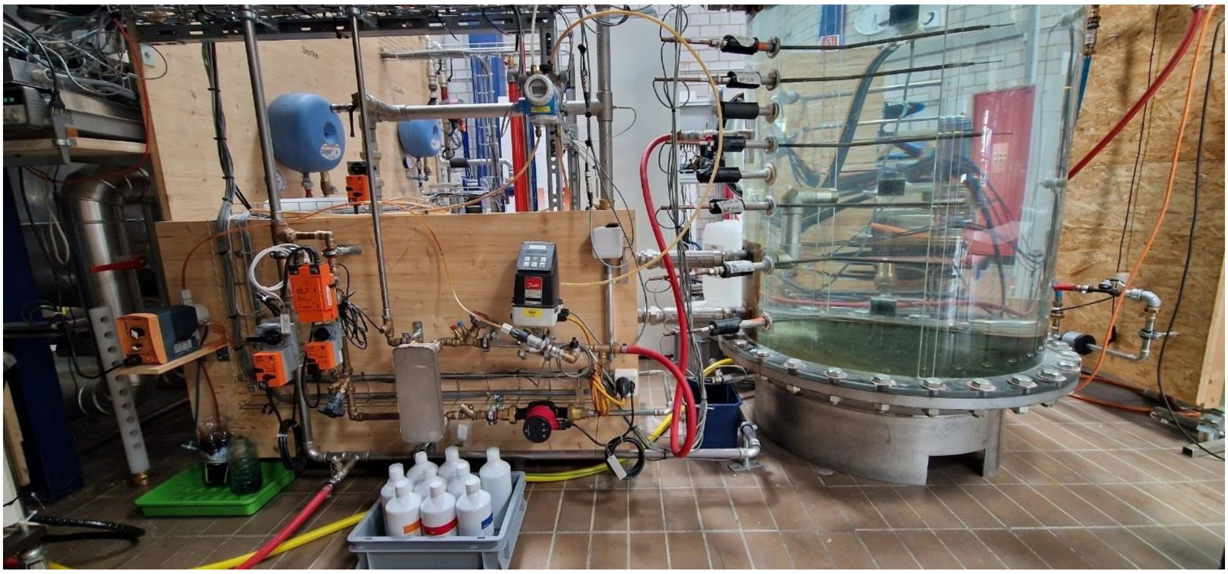

Figure 12: Hot Water System of the Test Setup

3.3.6 Suggestions for Improvement

During commissioning, various weaknesses in the test rig were identified. These relate in particular to venting, system contamination, measurement technology, and the accessibility of individual components.

Ventilation is a key issue. During operation, air accumulated in the system, making stable operation difficult. For future work, therefore, care should be taken to position the vent valves appropriately as early as the planning phase. Vent valves should preferably be located at high hydraulic points and in areas with low flow velocity. Figure 13 schematically shows the locations where additional venting options would be beneficial.

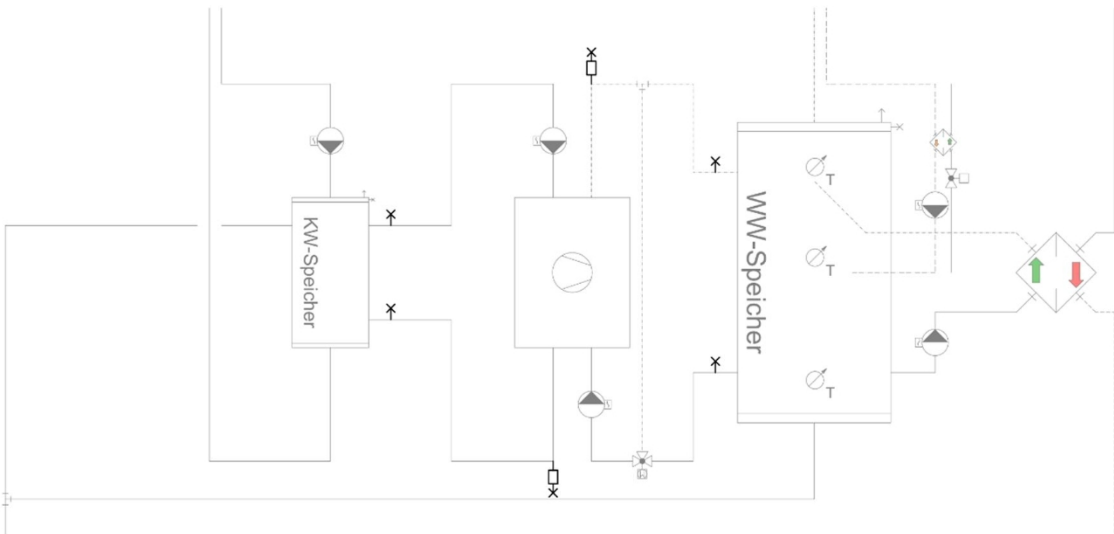

Figure 13 Schematic representation of recommended venting points

It was also determined that the Huba-Control flowmeters used are sensitive to mechanical vibrations. Vibrations can lead to unstable flow rate signals and measurement deviations. A specific critical flow velocity was not determined in this study. The observation therefore does not refer to a universally applicable threshold value, but rather to operating conditions with visibly increased vibrations or unstable measurement signals. In future modifications to the experimental setup, flow meters should therefore be mounted in a manner that minimizes vibrations as much as possible. In addition, it must be verified whether straight inlet and outlet runs can be maintained in accordance with the manufacturer’s specifications. Figure 14 shows the recommended principle for the arrangement of the flow measurement.

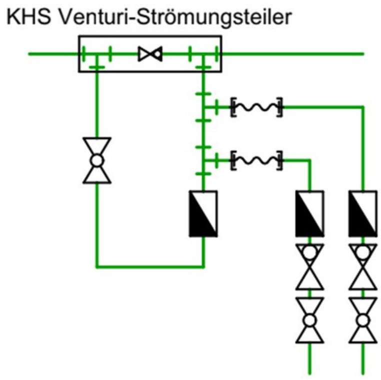

Figure 14 Recommended arrangement of the flow measurement with low-vibration mounting

System fouling also affected operation. In particular, in the area of the plate heat exchanger, deposits can reduce the volumetric flow rate and thereby limit the operating range. For future studies, additional filter elements, shut-off valves, and backflushing capabilities are therefore recommended. This can simplify maintenance and allow the test rig to be returned to a defined operating state more quickly. Figure 15 illustrates the principle of a more maintenance-friendly integration of the plate heat exchanger.

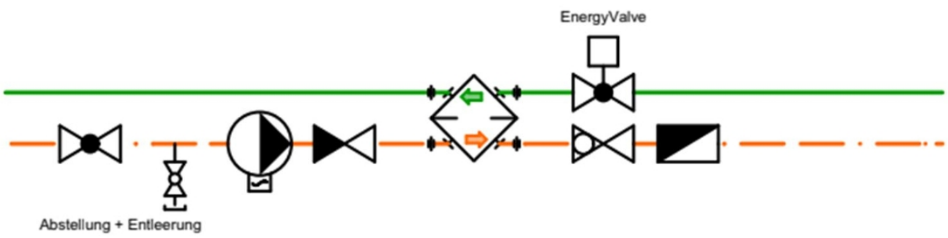

Figure 15: Principle for maintenance and backflushing of the plate heat exchanger

Before recommissioning, a revised operating concept for the heat pump should be defined in collaboration with the Institute of Mechanical Engineering. In particular, the source temperature, sink temperature, refrigerant pressures, compressor operation, and return flow maintenance should be taken into account. In addition, electrical performance measurements are recommended so that the heat pump's COP can be determined in future measurements. Figure 16 shows the relevant measurement and operating parameters for this purpose.

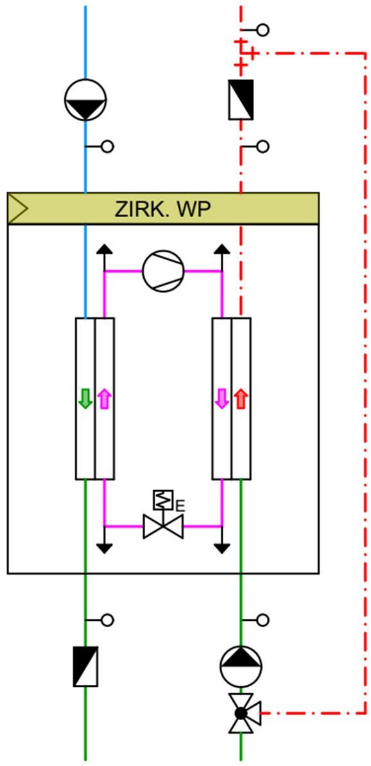

Figure 16 Relevant operating parameters of the heat pump

The most important optimization points are summarized in Table 13:

Table 13: Test Bench Optimization

Area	Observation	Recommended Action
Ventilation	Air pockets during operation	Additional vents at appropriate high points
Contamination	Reduced flow rate in the circulation loop	Install filter elements and provide for backwashing
Measurement technology	Sensitive flow measurement in the presence of vibrations	Install the flowmeter in a vibration-free location
Maintenance	Limited accessibility to individual components	Additional shut-off valves and detachable connections
Heat pump	Compressor failure	Review the operating concept with IME
Efficiency measurement	No electrical power measurement available	Add current measurement

3.4 Discussion: Test setup “ ” Circulation Heat Pump

The test setup illustrates the relationships described in Chapter 2 between the heat source, heat sink, storage tank management, and hydraulic decoupling. The cold water system is used as the heat source, while the hot water storage tank serves as the heat sink. The energy coupling occurs via the heat pump without the cold and hot water systems being hydraulically connected to each other.

The hot water storage tank was chosen as the interface to minimize the impact on hot water circulation. This allows the flow rate in the heat pump circuit to be adjusted independently of the existing circulation hydraulics. At the same time, heat is specifically introduced into the upper portion of the storage tank, thereby accounting for the importance of storage stratification in the experimental setup, as described in Chapter 2.

The setup illustrates which design constraints are essential for an experimental investigation. These include, in particular, reliable venting, robust flow measurement, suitable filter elements, adequate shut-off options, and good accessibility for maintenance and commissioning. These factors influence the measurement quality and operational safety of the test rig.

The test rig thus forms the basis for the subsequent test measurements in Chapter 4. An evaluation of actual operating behavior, storage stability, and energy performance is conducted only on the basis of the measurement data. Furthermore, a final energy assessment is possible only if the heat pump operates reliably and electrical power measurements are included.

4 Commissioning and : Test Measurements

4.1 Introduction: Commissioning and Test Measurements

This bachelor's thesis experimentally investigates the behavior of a circulation heat pump between a cold-water and a hot-water system. The goal of the measurements is to analyze the system's temperature-related and hydraulic behavior under various operating conditions. Particular focus is placed on the effects on storage stratification, the use of the cold-water system as a heat source, and the compensation for circulation heat losses.

The measurement campaign was planned to include various operating conditions, storage tank temperatures, and water withdrawal scenarios. However, due to a failure of the heat pump, it was not possible to evaluate complete measurement series. The evaluation is therefore based on the commissioning phase, the test measurements, and a longer-term measurement taken prior to the heat pump failure.

Despite the limited data available, qualitative insights into the system's behavior can be derived. These include, in particular, findings regarding hydraulic integration, storage stability, the control strategy, and weaknesses in the test setup. These insights serve as the basis for evaluating the test setup and for deriving optimization measures for future work.

4.2 Test Measurement Methodology

The measurements are based on the measurement protocol specified in Appendix B.1. This document defines the planned measurement series, the test sequence, the measured variables, and the termination and stability criteria. The test plan calls for investigations at storage tank temperatures of 50 °C and 60 °C, as well as various hot and cold water withdrawals.

Data acquisition is performed using LabVIEW and LabIGE. In particular, temperatures, flow rates, and storage tank states are recorded. From this data, the heat pump's supply and return temperatures, the temperature distribution within the storage tank, and the behavior of the cold water system can be evaluated. Additionally, energy balances on the source and sink sides are calculated from the measured variables.

Due to the failure of the heat pump, the planned measurement series could not be completed in full. The evaluation is therefore limited to the commissioning phase, the test measurements, and the existing extended measurement period prior to the heat pump failure. A detailed quantitative efficiency assessment is not performed, as no complete comparative measurements and no reliable electrical power measurements are available.

The evaluation focuses on the following points:

- hydraulic behavior of the test rig
- Temperature behavior in the chilled water system
- Heat input into the hot water storage tank
- Stability of the storage stratification
- Function of the control system
- Limitations of the test setup

Based on the available measurement data, it is not possible to make statements regarding the annual performance factor, long-term stability, or the overall energy efficiency of the system. These issues are additionally addressed theoretically in the calculation model in Chapter 5.

4.2.1 Definition of the “ ” draw-off conditions

The test measurements do not represent a complete daily withdrawal profile. Instead, defined draw-off conditions with a constant flow rate are used. This simplification was deliberately chosen because the focus of the study is not on a detailed representation of user behavior, but rather on the operational behavior of the circulation heat pump, its influence on the hot water storage tank, and heat extraction from the cold water system.

A real-world draw-off profile consists of short-term, highly fluctuating draws and can only be reproduced to a limited extent in the laboratory. However, comparable boundary conditions are crucial for evaluating the test setup. Constant draw-off conditions enable reproducible operating conditions and facilitate the analysis of temperature curves, flow rates, and energy flows. Short-term peak profiles are therefore not modeled separately.

The selected draw conditions are based on the hourly peak as defined in SIA 385/2. For the residential building under consideration, which accommodates 16 people, this results in an hourly peak factor of 38%, corresponding to an hourly value of approximately 212 l. For hot water withdrawal, a flow rate of 0.1 l/s is assumed for 35 minutes. This results in a hot water withdrawal of 210 l per measurement test. For cold water withdrawal, a flow rate of 0.2 l/s is assumed for the same period. This value is based on the assumed ratio of hot water to cold water demand of approximately 1:2. This results in a cold water withdrawal of 420 l per measurement test.

The selected flow rates therefore do not represent a complete simulation of a daily profile, but rather defined load cases for the laboratory test. This allows the effects of water withdrawal on the source, sink, and storage status to be specifically investigated.

4.3 Results of Commissioning and Test Measurements

During commissioning, the test rig was successfully operated hydraulically. After initial adjustments, the heat pump’s control system functioned stably and enabled continuous operation of the system over an extended period. The control settings originally selected initially proved to be too sluggish. However, after adjusting the control parameters, the heat pump exhibited stable and predictable operating behavior.

The measurements conducted reached steady-state operating conditions over extended periods. In particular, stable thermal stratification was observed within the hot water storage tank. The heat pump was able to maintain a stable temperature in the upper storage zone without significant mixing of the water within the tank.

In addition, a significant cooling of the chilled water system was observed. This confirms the heat pump’s function as a heat extraction system within the chilled water network. At the same time, the storage tank charging behavior was comparable to—or in some cases improved upon—that observed during operation without an active circulation heat pump. However, it should be noted that the capacities of the systems involved were not fully coordinated with one another.

The flow rates corresponded to the calculated design values. The hydraulic integration of the heat pump proved to be functional and plausible. Additionally, maintaining a high return flow rate on the sink side proved beneficial, as this helped support the stability of the storage tank stratification.

After approximately 18 hours of operation, the first problems arose with the heat pump. After about 20 hours, the heat pump—or more specifically, the compressor—failed completely. The exact cause could not be conclusively identified within the available project timeframe.

4.4 Qualitative Evaluation of the System Behavior

Due to the failure of the heat pump, the available measurement data do not permit a complete quantitative evaluation of the system. In particular, no reliable conclusions can be drawn regarding efficiency, COP, annual performance factor, or long-term behavior, as there are no complete measurement series, no repeat measurements, and no systematic variation in operating conditions. The evaluation in this chapter is therefore intentionally qualitative. The quantitative annual analysis is covered in addition in the calculation model in Chapter 5.

The qualitative statements are based on the commissioning, the test measurements, and the existing extended measurement period prior to the heat pump failure. The focus is on storage behavior, stratification within the storage tank, the duration between charging windows, and the supply and return temperatures of the heat pump on the source and sink sides.

4.4.1 Storage Behavior with and without a Circulation Heat Pump

To evaluate storage behavior, measurements taken with an active circulation heat pump are compared with a reference measurement taken without the heat pump in operation. The focus is not on a statistically reliable quantitative evaluation, but rather on a qualitative assessment of the upper storage zone, storage stratification, and charging behavior. Since complete repeat measurements are not available and the initial conditions are not entirely identical, the comparison should be understood as a qualitative assessment.

Figure 17 shows the temperature curve of the hot water storage tank without an active circulation heat pump. The measurement shows a reference operation in which the upper storage zone continuously cools down due to circulation losses. As soon as the activation temperature for the main charging cycle is reached, the storage tank is recharged. The charging windows for the main charging cycle are visible during the measurement period shown. This allows for an assessment of the cooling of the upper storage zone between two charging windows.

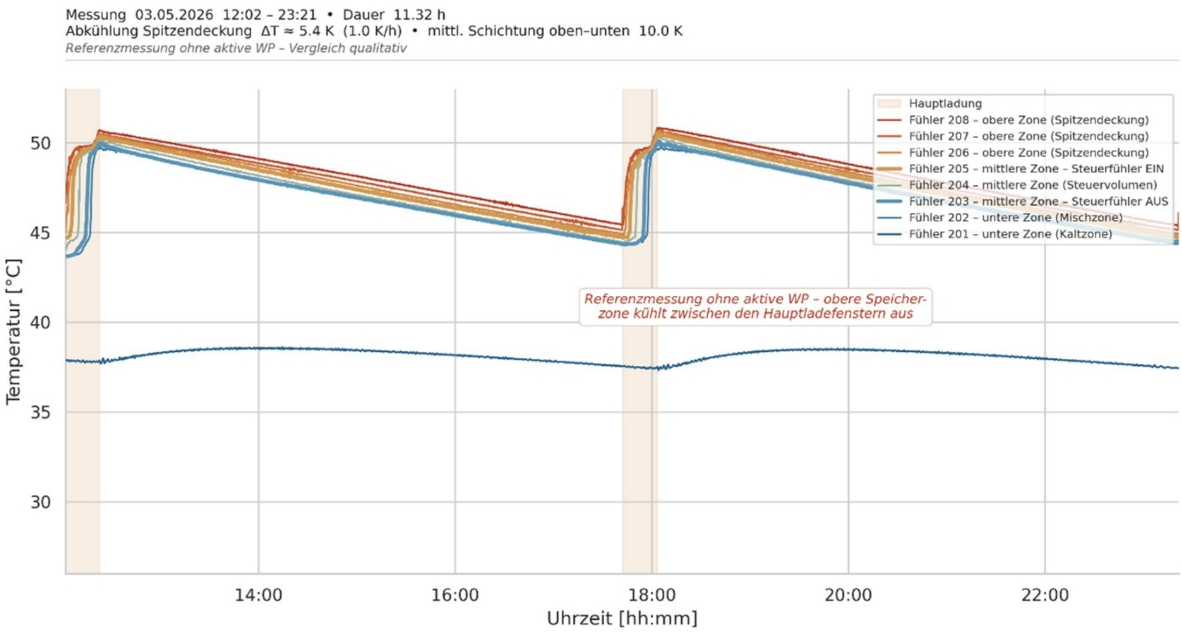

Figure 17 Temperature profile of the hot water storage tank without an active circulation heat pump

In reference operation without a circulation heat pump, the cooling of the upper storage zone during the period under consideration amounts to approximately 5.4 K. This results in an average cooling rate of about 1.0 K/h. During the measurement, the upper storage sensors consistently show higher temperatures than the lower storage areas. The thermal stratification thus remains essentially intact, but the upper storage zone continuously loses temperature until the main charging cycle is required again.

Figure 18 illustrates the corresponding operation with the circulation heat pump active. The heat pump draws water from the lower storage area, heats it, and then returns it to the upper storage area. The measurement interval shown depicts a nearly complete charging cycle; however, the start of the main charging phase is only partially visible. The figure is therefore not used as a fully reliable cycle comparison, but rather for a qualitative assessment of the storage tank support provided by the heat pump.

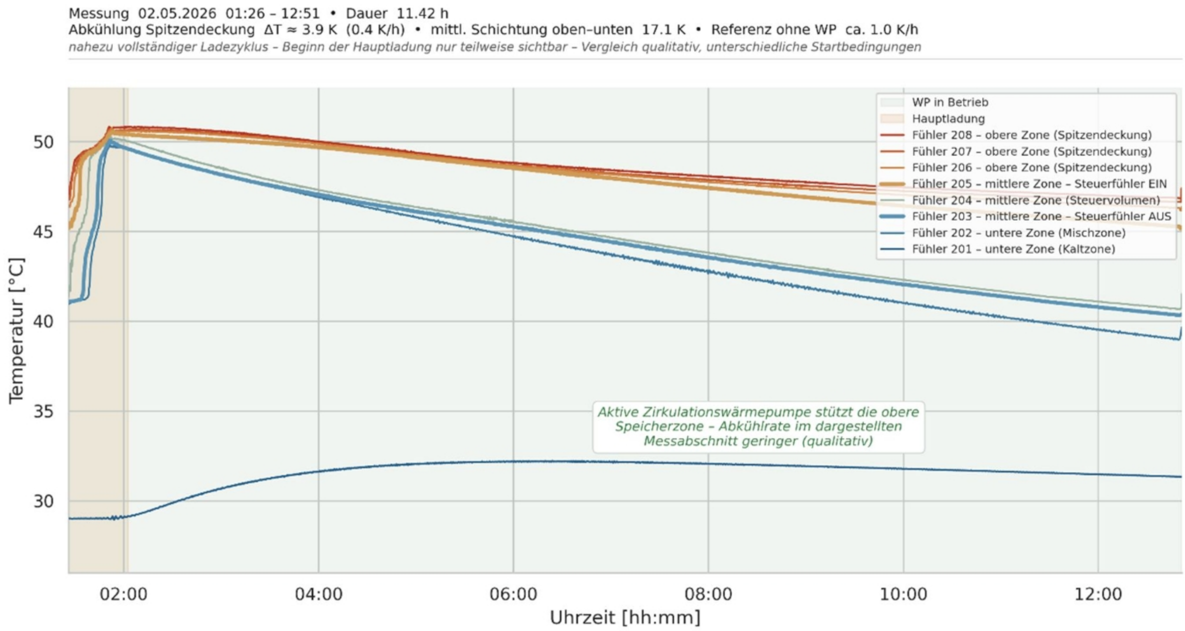

Figure 18 Temperature profile of the hot water storage tank with an active circulation heat pump

With the circulation heat pump active, the cooling of the upper storage zone in the measurement segment shown is approximately 3.9 K. The average cooling rate is about 0.4 K/h. Compared to the reference measurement, the cooling of the upper storage zone is thus less pronounced. This suggests that the heat input from the circulation heat pump supports the top layer during the period under consideration.

The measurements thus demonstrate qualitatively that the active circulation heat pump can slow the cooling of the upper storage zone. At the same time, the stratification of the storage tank is largely maintained. The upper temperature sensors continue to show higher temperatures than the lower storage zones. No complete mixing of the storage tank is observable in the measurement period under consideration.

When interpreting the results, it must be taken into account that the measurements are not entirely equivalent comparative measurements. The initial conditions and the visible charging profile differ between the two measurements. Furthermore, there is no systematic repetition of the operating conditions. The results therefore do not show a statistically significant increase in efficiency, but rather a qualitative trend: The heat pump can support the upper storage zone and thereby positively influence the storage tank's charging behavior.

To contextualize the observed storage response, the most important measurement parameters are summarized in Table 14. The table serves as a qualitative comparison and not as a conclusive quantitative evaluation.

Table 14 Comparison of Storage Behavior with and without a Circulation Heat Pump

Parameter	Without circulation heat pump	With circulation heat pump	Evaluation
Measurement duration	approx. 11.3 h	approx. 11.4 h	Similar observation duration
Charging cycle	Similar observation duration	Nearly complete charging cycle; start of final charge only partially visible	Qualitative comparison
Cooling of the upper storage zone	approx. 5.4 K	approx. 3.9 K	slightly lower with heat pump
Average cooling rate in the upper storage zone	approx. 1.0 K/h	approx. 0.4 K/h	reduced with WP
Layering in the storage tank	Upper zone warmer than lower zone	Upper zone warmer than lower zone	Stratificatio n remains intact
Statement regarding the next main charge	Can be assessed in the reference curve	Can only be assessed to a limited extent	No definitive cycle assessment

The comparison shows that the active circulation heat pump reduces the cooling rate of the upper storage zone in the section under consideration. However, due to limited comparability, a reliable statement regarding the extension of the time until the next main charge is only possible in qualitative terms.

4.4.2 Flow and return temperatures of the heat pump

In addition to storage behavior, the supply and return temperatures of the heat pump on the source and sink sides are evaluated. These parameters indicate whether the heat pump extracts heat from the cold-water system and transfers it to the hot-water system. The evaluation serves as a qualitative functional check of the heat pump during the operating period under consideration.

On the source side, the temperature difference between the heat pump’s inlet and outlet is examined. During operation, the outlet temperature on the source side is lower than the inlet temperature. This indicates that heat is being extracted from the cold water system. During the stable operating period under consideration, the average temperature difference on the source side is approximately 2.0 K.

On the sink side, the opposite behavior is observed. The heat pump’s outlet temperature is higher than the inlet temperature. This indicates that heat is being transferred to the hot water system. The average temperature difference on the sink side is approximately 3.4 K during the operating period under consideration.

To verify this heat transfer, the supply and return temperatures of the heat pump on the source and sink sides are plotted together. Figure 19 shows the temperature profile during the operating period under consideration. The graph is divided into the sink side and source side so that heat extraction from the chilled water system and heat transfer to the hot water system can be evaluated separately.

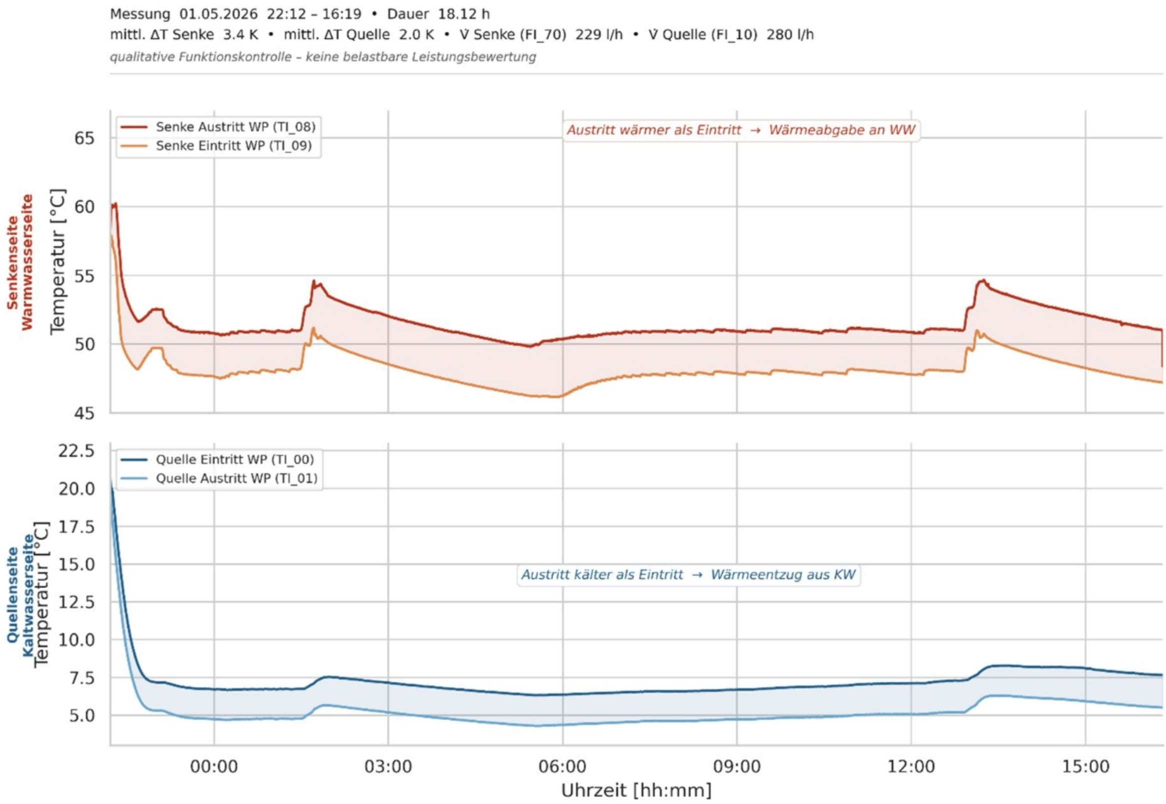

Figure 19: Supply and return temperatures of the heat pump on the source and sink sides

The temperature profiles confirm the expected behavior of the heat pump. On the source side, the outlet is colder than the inlet, which illustrates the heat extraction from the chilled water system. On the sink side, the outlet is warmer than the inlet, which demonstrates the heat transfer to the hot water system. The flow rates in the section under consideration are approximately 350 l/h on the source side and approximately 229 l/h on the sink side. Thus, the temperature differences can be interpreted as hydraulically plausible.

In addition to the temperature differences, Figure 20 presents a qualitative comparison of the calculated source and sink capacities. This analysis is based on the measured temperature differences and flow rates. It serves to plausibly contextualize the heat transfer via the heat pump.

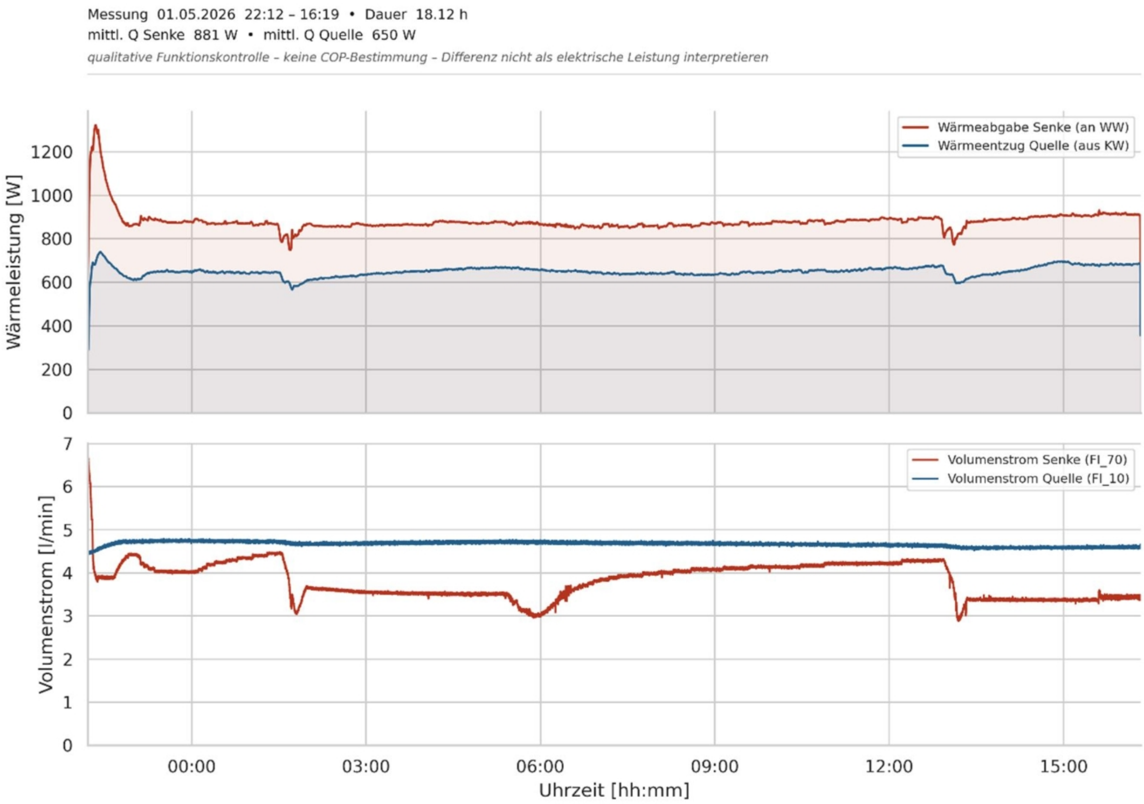

Figure 20 Qualitative Comparison of the Calculated Source and Sink Power

The calculated average source power in the section under consideration is approximately 650 W. The calculated average sink power is approximately 881 W. The sink power is thus higher than the source power, which generally corresponds to the expected behavior of a heat pump, since on the sink side, in addition to the heat extracted from the source, the electrical compressor power is also dissipated as heat.

No COP is determined from this comparison of power ratings. The difference between source and sink power is approximately 231 W and should not be interpreted as electrical compressor power, as the measurement limits for this are not sufficiently defined. Possible causes for the small difference include measurement uncertainties in the temperature sensors, flow rate measurements, different sensor locations, delays in the water circuit, and heat losses in piping and components.

Figure 20 is therefore used exclusively as a qualitative plausibility check. It shows that, during the operating period under consideration, heat is extracted on the source side and heat is released on the sink side. A reliable efficiency assessment, a true COP, or an annual performance factor cannot be derived from this.

To assess the heat pump’s operation, the most important measurement parameters are summarized in Table 15. The table serves as a qualitative functional check and does not replace a complete performance or efficiency calculation.

Table 15: Heat Pump Parameters During the Operating Period Under Consideration

Parameter	Source side	Load side
Qualitative Function	Heat Extraction from Cold Water	Heat transfer to hot water
Temperature behavior	Outlet colder than inlet	Outlet warmer than inlet
Average temperature difference	approx. 2.0 K	approx. 3.4 K
Average flow rate	approx. 350 l/h	approx. 229 l/h
Calculated average power	approx. 650 W	approx. 881 W
Rating	Visible heat dissipation	Heat release visible

Based on the measured supply and return temperatures, the flow rates, and the qualitative performance comparison, the basic function of the heat pump can be confirmed. The heat pump extracts heat from the cold water system and transfers it to the hot water system. However, due to the lack of electrical power measurements and the limited data set, the evaluation is restricted to a qualitative functional check.

4.4.3 Classification of the Control System

In this study, the control of the circulation heat pump is understood as a higher-level system control. This does not primarily refer to the internal compressor control, but rather to the activation and deactivation of the heat pump depending on the storage tank and source temperatures. The possible control approaches are described in Chapter 2.3.8.

The test measurements focus on hot-water-oriented activation. The heat pump is activated when the upper storage zone is capable of absorbing heat. This prioritizes the stabilization of the hot-water storage tank. A purely cold-water-oriented activation is not fully investigated within the scope of the available measurement data.

After adjusting the controller settings, the system exhibits predictable operating behavior. The heat pump responds to the storage tank’s state and transfers heat to the upper storage zone. At the same time, the measurements show that the control system is highly dependent on stable hydraulic conditions. Air pockets, contamination, or unstable flow rates can impair control quality and operational reliability.

4.4.4 Limitations of the Analysis

The available measurement data illustrate the basic system behavior but are insufficient for a conclusive energy assessment. Due to the heat pump failure, it was not possible to conduct complete measurement series, repeat measurements, or systematic variations in operating conditions.

To transparently assess the significance of the available measurement data, the research questions examined are summarized below. Table 16 shows which conclusions can be qualitatively derived from the measurements and which assessments cannot be reliably made due to the lack of complete measurement series.

Table 16: Validity of the Available Measurement Data

Research Question	Assessment Using Available Data
Hydraulic feasibility	Qualitatively possible
Storage behavior	Can be assessed qualitatively
Stratification stability	Can be qualitatively assessed
Heat extraction from cold water	Visible via temperature difference
Heat transfer to hot water	Visible due to temperature difference
COP / Efficiency	Not applicable
Annual performance factor	Cannot be determined
Long-term performance	Cannot be determined

The measurements thus show that the hydraulic integration and the basic principle of the circulation heat pump can be qualitatively evaluated. However, further measurement series with a fully functional heat pump, electrical power measurement, and repeatable operating conditions are required for a reliable energy assessment.

4.5 Limitations of the Study

Due to technical problems, the study has various limitations that must be taken into account when evaluating the results. Because of the heat pump failure, the originally planned measurement campaign could not be carried out in full. As a result, only limited measurement data is available for evaluation.

Additionally, no repeat measurements could be performed. This complicates the assessment of reproducibility as well as the evaluation of potential measurement uncertainties. A statistical analysis of the operating behavior is therefore not meaningful.

Furthermore, no complete electrical power measurements of the heat pump were available. Consequently, the efficiency of the refrigeration cycle, including the compressor, cannot be reliably determined. There are also no documented manufacturer performance curves for the laboratory heat pump used. Although individual measured values from BAT_G_25_09 show COP values that depend on operating conditions, these are strongly influenced by source temperature, sink temperature, flow rate, and operating mode and cannot be used to represent the general efficiency of the refrigeration cycle (Degen & Kunz, 2025).

Conclusions regarding efficiency are therefore not derived from the test measurements in this study, but are only theoretically estimated using the computational model. Conclusions regarding the annual performance factor, the actual COP, or the long-term behavior of the system are not possible based on the available measurement data.

This thesis therefore deliberately focuses on the qualitative evaluation of system behavior as well as on the insights gained from commissioning and the test measurements.

4.6 Suggestions for Future Work

For future work, it is recommended to first ensure the operational reliability of the heat pump. In particular, the causes of the compressor failure should be investigated in detail in collaboration with the Institute of Mechanical Engineering. The control strategy, operating pressures, and thermal operating conditions of the heat pump appear to be particularly relevant in this regard.

In addition, the test rig should be further optimized hydraulically. This includes, in particular, improvements to the venting system, additional filter elements, and the ability to backflush the system. This could reduce contamination within the plate heat exchanger and improve hydraulic stability.

For future measurement campaigns, it is also recommended to integrate electrical power measurements and temperature measurements within the refrigerant circuit. This would allow for more reliable conclusions regarding the efficiency and thermodynamic behavior of the heat pump.

Furthermore, an investigation of alternative heat sources and sinks appears worthwhile. In particular, using the utility room as an alternative heat sink and the targeted thermal overcharging of the hot water storage tank have the potential to improve the operating strategy.

In addition, future work should include longer measurement campaigns under reproducible operating conditions in order to draw conclusions regarding the system’s long-term stability, efficiency, and hygienic performance.

4.7 Discussion: Commissioning and “ ” Test Measurements

Despite the limited data available, this study yielded important insights into the hydraulic integration and operational behavior of a circulation heat pump. The results show that the basic concept is technically feasible and that, in particular, the use of the cold-water system as a heat source holds potential.

The qualitative evaluation suggests that the heat pump can help stabilize the stratification in the storage tank and reduce the load on the main heating system. At the same time, it became apparent that

control and hydraulic stability, in particular, are crucial for reliable operation.

The problems encountered also illustrate that experimental prototype systems can be sensitive to hydraulic, control-related, and thermodynamic boundary conditions. In particular, air bleeding, system fouling, and the operational reliability of the heat pump have a significant influence on the functioning of the overall system.

This thesis therefore does not provide a complete energy assessment of the entire system, but rather a qualitative functional evaluation and a basis for further measurement campaigns.

5 Simulation and : Computational Model

5.1 Introduction: Simulation and the “ ” Computational Model

To complement the experimental investigations, a simplified energy storage model was developed. The goal of the model is to theoretically estimate the operational behavior of a circulation heat pump in interaction with the chilled water network, the hot water storage tank, and hot water demand.

The model does not represent the hot water storage tank in detailed layers but instead calculates its total energy content. This allows for an analysis of the extent to which the circulation heat pump extracts heat from the chilled water network and transfers it to the hot water storage tank. At the same time, the study examines when an additional main charge of the storage tank becomes necessary and how the cold water temperature changes as a result of the heat pump’s operation.

Due to the failure of the heat pump, the model results could not be validated with complete measurement series. The model therefore does not serve as an exact representation of actual operation, but rather as a theoretical estimate of the energy potential and possible operating limits.

5.2 Methodology: Simulation and “ ” Calculation Model

The calculation model was developed in an Excel workbook and operates using hourly time steps over a full year. A finer temporal resolution was not implemented within the scope of this work. The hourly time step was chosen because the operating temperatures, draw-off profiles, and energy balance parameters used are processed as hourly values, thereby enabling a clear annual analysis. Short-term dynamic processes, such as individual draw-off peaks or the cycling behavior of the heat pump, are therefore not represented in detail.

The analysis is based on the input parameters of the representative building, the calculated circulation heat losses, the operating temperatures of the chilled water, and defined draw-off profiles for hot and cold water.

The model is based on simplified boundary conditions. The circulation losses in the hot water distribution system are assumed to be constant throughout the year. Consequently, seasonal fluctuations in ambient temperatures in mechanical rooms, shafts, or installation zones are not dynamically represented. Hot water demand is also not adjusted seasonally in the model but is calculated based on a constant usage profile. Possible differences between summer and winter operation are therefore not taken into account. These simplifications allow for a clear annual energy analysis but limit the model’s ability to reflect real dynamic operating conditions.

The most important input parameters are defined in the “Inputs” spreadsheet. These include, among other things, circulation losses in the hot water distribution system, storage losses, discharge losses, storage volume, cold water volume, temperature limits, and the maximum heating capacity of the heat pump. The hot water side is modeled with a control volume of 400 L. The setpoint temperature for the hot water storage tank is set to 58 °C. Additionally, a maximum overcharge temperature of 70 °C can be accounted for in the model.

On the chilled water side, a 50-liter chilled water storage tank and a 50-liter piping volume are taken into account. The heat pump may only be operated if the chilled water temperature is above the defined lower operating limit. A setpoint temperature of 4 °C is used for the chilled water.

The draw profiles are specified in the "Draw Profiles" spreadsheet. From these, the model calculates hourly hot and cold water withdrawals. The operating temperatures of the cold water are stored in the "Operating Temperature" spreadsheet and serve as time-dependent boundary conditions for the source temperature.

The actual calculation takes place in the "Simulation" worksheet. There, the storage tank status, the cold water temperature, heat pump operation, heat transfer to the hot water storage tank, heat extraction from the cold water, and the required main charge are calculated for each hour. The results are then summarized in the "Evaluation" worksheet and presented graphically in the "Diagrams" worksheet.

The maximum heat transfer to the hot water storage tank used in the calculation model is 500 W. This value describes the modeled sink side and is not directly equivalent to the cooling capacity of approximately 600 W of the heat pump used in the laboratory. The 600 W refers to the source side of the laboratory heat pump, while the 500 W in the model is set as the limited heat transfer to the storage tank.

The most important input parameters of the model are shown in Table 17.

Table 17 Input Parameters of the Thermal Storage Model

Parameter	Value	Unit
Hot water circulation loss	484.5	W
Hot water storage loss	1.5	kWh/d
Hot Water Emission Loss	0.8	kWh/day
Target temperature of hot water storage tank	58	°C
Maximum overcharge temperature	70	°C
Control volume	400	l
Cold water storage tank capacity	50	l
Volume of cold water distribution system	50	l
Target temperature of cold water	4	°C
Reference circulation gain for cold water	329.67	W
Max. heat transfer to DHW storage tank in the model	500	W
Assumed Carnot efficiency of the heat pump	0.45	-
Daily hot water draw	576	l/d
Daily cold water consumption	1,242	l/d
Time step	1	h

The results are then analyzed graphically and evaluated qualitatively.

5.3 Structure and Calculation of the “ ” Simulation Model

The model is based on an energy balance analysis of the hot water storage tank. The storage tank is not calculated based on individual temperature layers, but is described by its energy content in Wh. The full storage state corresponds to the amount of energy resulting from the storage volume, the setpoint temperature of 58 °C, and the cold inlet temperature. Additionally, a maximum storage state is calculated for a possible overcharge up to 70 °C.

At each time step, the current hot water demand is first determined. This consists of the circulation loss, the storage and discharge losses, and the draw-off energy. The draw-off energy is calculated from the hourly hot water withdrawal and the temperature difference between the hot water and the cold inlet water.

At the same time, the cold water side is balanced. The cold water temperature is determined by the available cold water volume, the cold water withdrawal, the ambient temperature, and the heat input from the environment or the piping network. Additionally, the amount of heat extracted from the cold water by the heat pump is taken into account.

The heat pump’s operation is controlled by simple operating conditions. The heat pump runs only when the cold water temperature is above the lower operating limit and the hot water storage tank can absorb heat. If the storage tank is fully charged or if the cold water is too cold, the heat pump is disabled in the model.

The heat output of the heat pump is limited to a maximum value. The heat released on the sink side is attributed to the hot water storage tank. The heat extracted on the source side is calculated from the heat released and the theoretical COP.

The COP is estimated in a simplified manner using the ideal Carnot COP and an assumed coefficient of performance. The Carnot COP describes the theoretically maximum possible COP as a function of the evaporation and condensation temperatures. Since real heat pumps do not achieve this ideal process, the Carnot COP is reduced by a coefficient of performance. In the model, a constant efficiency of 0.45 is used for this purpose. This allows the influence of the temperature difference between the source and the sink to be taken into account in a simplified manner. Since no electrical power measurements and no complete performance curves for the laboratory heat pump are available, the calculated COP is a theoretical reference value.

If the hot water storage tank falls below the defined minimum energy level, a main recharge is activated. This represents a conventional recharge of the storage tank and serves, in the model, to ensure the hot water supply. This allows us to investigate how often and to what extent the main recharge remains necessary despite the active circulation heat pump.

5.4 Sample Calculation for a Representative “ ” Building

The example calculation was performed using a representative multi-family residence. The model calculates the energy content of the hot water storage tank, the temperature of the cold water system, and the operation of the circulation heat pump on an hourly basis over a full year.

The operating temperature of the chilled water is used as a time-dependent boundary condition. The annual duration curve in Figure 21 shows that the operating temperature ranges between approximately 8 °C and 13 °C for a large part of the year. In some hours, lower values of about 5 °C and peak values of up to around 17 °C are reached. The comfort upper limit of 15 °C is exceeded during a significant portion of the year. As a result, there is usable potential for cooling the chilled water, particularly during warmer periods.

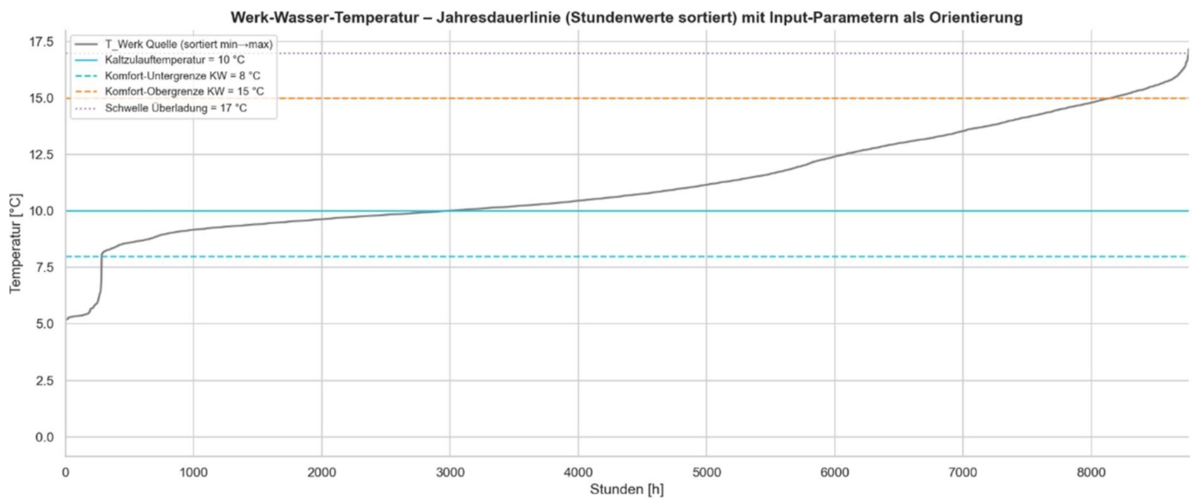

Figure 21 Annual duration curve of the plant temperature of the chilled water with relevant threshold values

Based on these operating temperatures, the model calculates the resulting temperature in the chilled water network. Figure 22 shows the trend of the calculated chilled water temperature as a daily average over the course of the year. The temperature rises continuously starting in spring and reaches its highest values in summer. In the fall, it drops again. The trend shows that, despite heat extraction by the heat pump, the chilled water network exhibits seasonal fluctuations, with elevated temperatures occurring particularly in the summer.

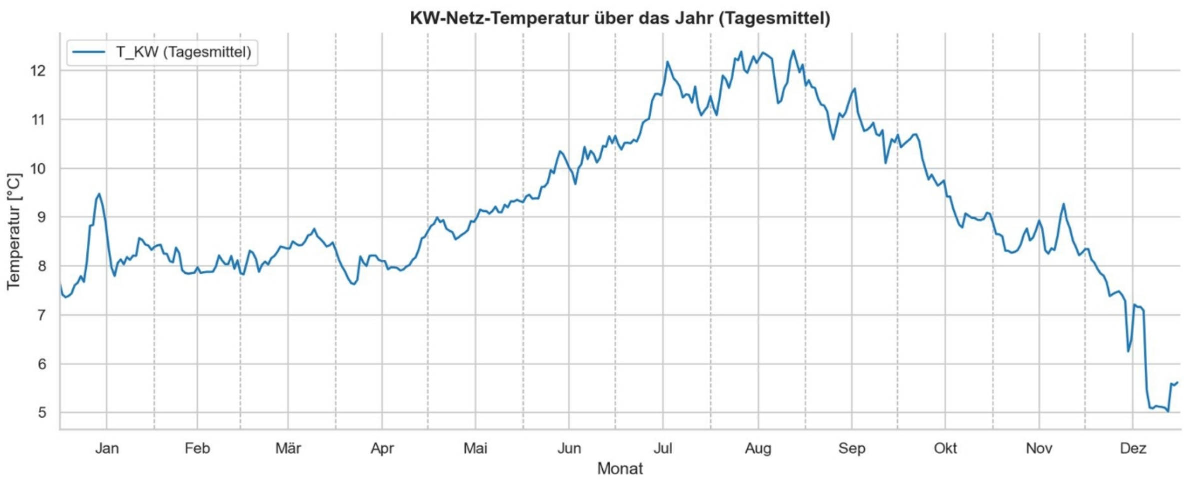

Figure 22: Cold-water network temperature over the course of the year as a daily average

The annual duration curve for the chilled water network temperature is shown in Figure 23. It indicates the number of hours during which specific temperature ranges occur in the calculated chilled water network. The temperature never falls below the lower operating limit of the heat pump or the setpoint of 4 °C. The comfort upper limit of 15 °C is also never exceeded in the model under consideration. The threshold for storage tank overcharge of 17 °C is not reached.

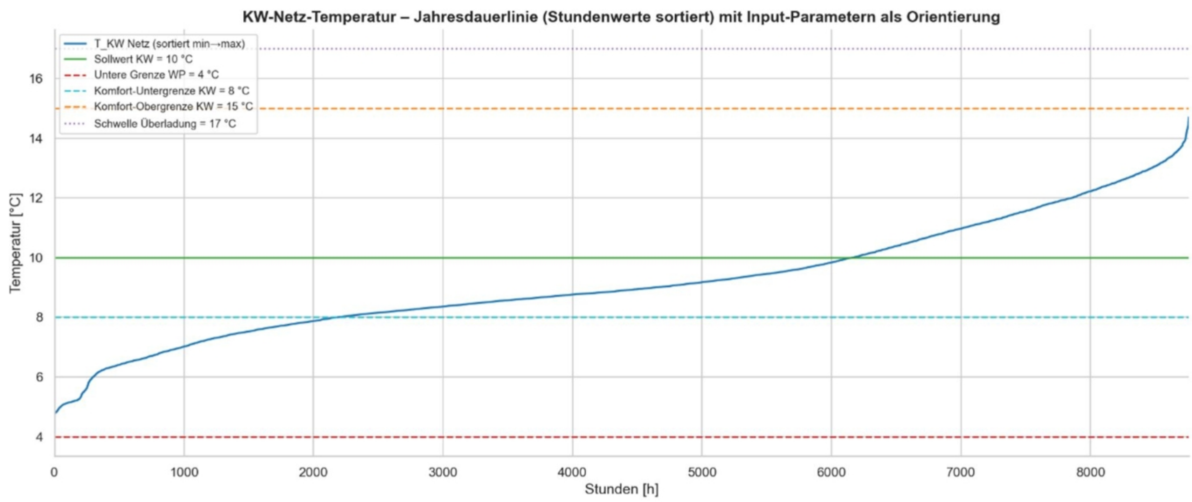

Figure 23 Annual duration curve of the chilled water network temperature with input parameters

Figure 24 shows the daily number of hours during which the calculated chilled water temperature lies outside the defined comfort range of 8 °C to 15 °C. A distinction is made between excessively high chilled water temperatures above the comfort upper limit and excessively low chilled water temperatures below the comfort lower limit. In the model under consideration, the comfort upper limit of 15 °C is never exceeded. However, temperatures below 8 °C are calculated, particularly during the winter months and, to some extent, during the transitional seasons. The circulation heat pump thus prevents excessively high chilled water temperatures in the model but can occasionally lead to comfort restrictions due to chilled water that is too cold.

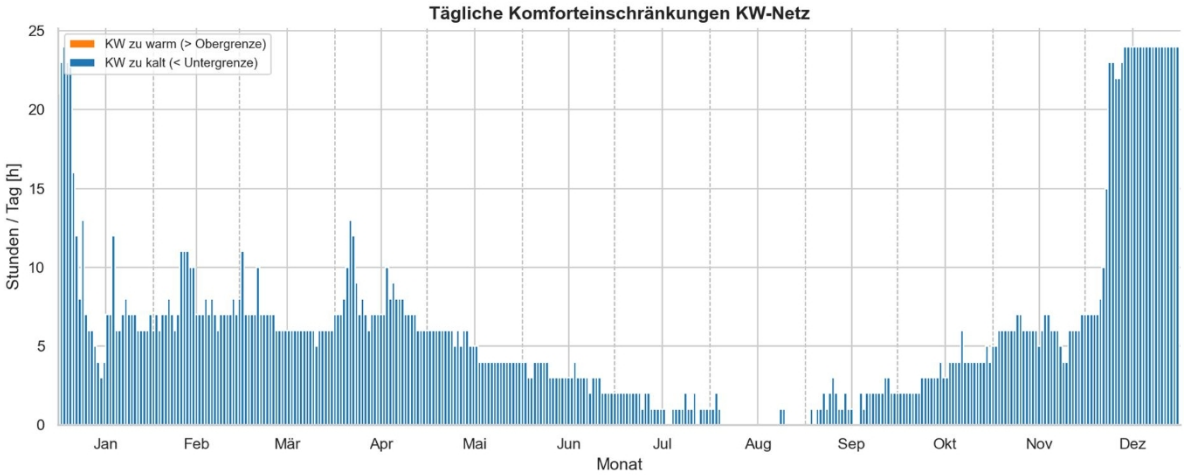

Figure 24: Daily hours outside the comfort range for cold water

When interpreting the comfort analysis, it should be noted that the range from 8 °C to 15 °C is used in this study as the comfort-related evaluation range for user perception. Temperatures below 8 °C are not considered hygienically critical but may be perceived as uncomfortably cold. Temperatures above 15 °C, on the other hand, are perceived as less refreshing and may also become more hygienically relevant if temperatures continue to rise. The model thus shows that heat removal from the cold water must be controlled not only with regard to the upper comfort limit but also with regard to a lower comfort limit.

When interpreting the model results, it must also be taken into account that, in reality, circulation losses depend significantly on the temperature difference between the heated pipe and the surrounding environment. Since ambient temperatures in utility rooms, shafts, and installation zones are not modeled as time-dependent in the model, seasonal changes in circulation losses are not accounted for.

Furthermore, hot water demand does not vary seasonally in the model. In real residential buildings, hot water consumption can fluctuate throughout the year. Literature values show that hot water consumption in residential buildings can decrease during the warmer months compared to the annual average. Lower hot water demand in the summer can reduce available storage capacity, while at the same time higher cold water and ambient temperatures can increase the source potential. The presented analysis of the cold water temperature should therefore be understood as a computational estimate based on the selected model assumptions.

In the model, the circulation heat pump is limited to a maximum heat output of 500 W on the sink side. Figure 25 shows the average daily output of the heat pump. The constant line at 500 W indicates that, in the model, the heat pump is limited by its maximum output during operation. The model thus does not simulate dynamic modulation of the heat pump but evaluates the system with a fixed maximum heat output to the hot-water storage tank.

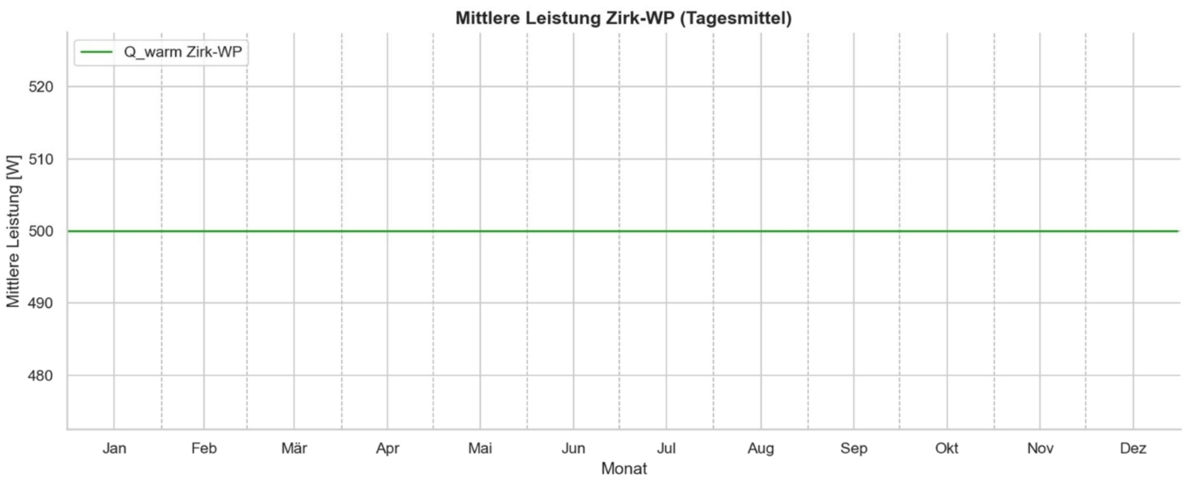

Figure 25 Average power of the circulation heat pump as a daily average

The annual analysis shows that the circulation heat pump in the model transfers 4,380 kWh/a of heat to the hot water storage tank. Of this, 2,947 kWh/a is drawn from the cold water system. The calculated electricity consumption of the heat pump is 1,434 kWh/a. This results in a theoretical annual coefficient of performance of approximately 3.06.

The main charging of the hot water storage tank remains necessary despite the circulation heat pump. In the model, the main charging system supplies 11,066 kWh/a and is active for 892 hours per year. This shows that the circulation heat pump does not replace the main heat generator but acts as a supplementary system to support the storage tank. The heat pump reduces the energy requirement of the main charging system but cannot guarantee the reliability of the hot water system's supply on its own.

The annual average cold water temperature is 9.2 °C. The lowest calculated temperature is 4.7 °C, and the highest is 14.7 °C. The comfort upper limit of 15 °C is therefore not exceeded for a single hour of the year. However, the comfort lower limit of 8 °C is sometimes fallen short of, particularly during the winter months and to some extent during the transitional seasons. It follows that, in the model under consideration, the heat pump sufficiently cools the cold water to prevent it from reaching excessively high temperatures; however, heat extraction must be controlled so that the cold water is not cooled unnecessarily far below the comfort lower limit.

At the same time, it should be noted that this result depends heavily on the specified heat pump capacity, the control strategy, and the selected model assumptions. With different capacity values or altered operating conditions, it cannot be guaranteed in every operating state that the chilled water temperature will remain within the comfort range at all times. The analysis therefore shows that a capacity-controlled or temperature-dependent heat pump is advisable in order to adapt heat extraction to the available source potential and the current chilled water temperature. At low cold-water temperatures, the heat pump must be reduced or shut down, while at higher cold-water temperatures, greater heat extraction may be appropriate. This allows the comfort range to be maintained more consistently and prevents unnecessary excessive cooling of the cold water.

The results show that the circulation heat pump operates most effectively when there is a simultaneous heat demand in the hot water storage tank and usable heat in the chilled water network. At low chilled water temperatures, the energy potential decreases. At higher chilled water temperatures, the heat pump's operating point improves because the temperature difference between the source and sink becomes smaller.

To assess energy efficiency, the theoretical electricity consumption of the circulation heat pump is compared with that of direct electric reheating. The calculated electricity demand of the heat pump is 1,434 kWh/a. Assuming an electricity price of 0.30 CHF/kWh, this results in annual electricity costs of approximately 430 CHF/a. If auxiliary energy for circulation pumps, control systems, and metering systems is also taken into account, the actual electricity demand is higher.

If the 4,380 kWh/a of heat output from the heat pump were provided directly by electricity—for example, via an electric heating element or an electric heating cable—the electricity costs would amount to approximately 1,310 CHF/a at the same electricity price. The calculated difference between the heat pump and direct electric reheating is thus approximately 880 CHF/year.

This comparison highlights the energy efficiency advantage of the heat pump over direct electric reheating. However, it does not constitute a complete economic analysis. A heat pump system incurs additional costs due to circulation pumps, control systems, measurement and safety devices, maintenance, and potential repairs to the refrigeration circuit. Furthermore, the upfront investment costs for a heat pump are higher than those for simple electric reheating.

An electric heating tape is technically simple to install and can deliver heat directly along the pipe. In terms of energy efficiency, however, direct electric reheating corresponds to a COP of approximately 1, since the electrical energy used is converted directly into heat. In contrast, the circulation heat pump additionally utilizes heat from the chilled water network. The economic benefit of the heat pump therefore depends largely on the actual annual coefficient of performance, the additional auxiliary energy, the maintenance costs, the service life, and the investment costs.

The heat pump is energetically advantageous; however, it must justify this advantage economically through sufficiently long operating times, stable efficiency, and low additional operating costs. A comprehensive economic analysis is not part of this study but would be necessary for subsequent product development or market launch.

The simplified comparison is shown in Table 18. The table considers only the electricity consumption for the actual reheating process. Investment costs, maintenance, auxiliary energy, servicing, and service life are not taken into account.

Table 18 Simplified Energy Assessment of Post-Heating

Variant	Electricity Consumption	Electricity costs at 0.30 CHF/kWh	Note
Circulation heat pump	1,434 kWh/year plus auxiliary energy	approx. 430 CHF/year plus auxiliary energy	uses heat from the cold water network
Direct electric reheating	4,380 kWh/year	approx. 1,310 CHF/year	Simple technology, COP ≈ 1
Calculated difference	**2,946 kWh/year**	**approx. 880 CHF/year**	**excluding investment, maintenance, and auxiliary energy**

The sample calculation thus shows that the circulation heat pump in the model can handle a significant portion of the reheating of the hot water storage tank. At the same time, the effect on the chilled water system depends heavily on the selected model assumptions. In the model under consideration, the calculated chilled water temperature remains entirely below the comfort upper limit of 15 °C. At the same time, temperatures occasionally fall below the comfort lower limit of 8 °C. The effect on the chilled water system must therefore be assessed not only in terms of excessively high temperatures but also in terms of potential overcooling. With different performance parameters, control strategies, or operating temperatures, both exceedances of the upper comfort limit and falls below the lower comfort limit may occur. The results should therefore be understood as an estimate of energy potential based on simplified model assumptions. A reliable assessment of actual operation requires measurement data from a fully functional heat pump, as well as consideration of seasonal usage patterns and environmental influences.

5.5 Discussion: Simulation and “ ” Calculation Model

The calculation model shows that the chilled water network can, in principle, be used as a heat source for a circulation heat pump. The heat pump extracts heat from the chilled water system and uses it to supplement the hot water storage tank. This reduces the main charge of the storage tank but does not completely replace it.

The results also show that the efficiency of the circulation heat pump depends heavily on the cold water temperature. The annual duration curves show that the operating temperature exceeds the upper comfort limit during part of the year. After heat extraction by the heat pump, the calculated cold water temperature in the model under consideration remains entirely below the upper comfort limit of 15 °C. The heat pump thus sufficiently reduces the thermal load on the cold water with regard to excessively high temperatures. At the same time, the analysis of comfort limitations shows that the cold water temperature falls below the lower comfort limit of 8 °C, particularly in winter and to some extent during transitional seasons. The operation of the heat pump must therefore be controlled by both an upper temperature limit and a lower comfort or operating limit. This result depends heavily on the specified heat pump capacity, the control strategy, and the simplified boundary conditions. For other buildings, operating temperatures, or performance metrics, no universally valid guarantee of compliance with the comfort range can be derived from this.

The calculated annual coefficient of performance (COP) of approximately 3.06 should be understood as a theoretical model value. It is based on a simplified COP approach using the Carnot efficiency and not on measured electrical power consumption. Real-world operating conditions such as compressor starts, part-load behavior, pressure losses, control deviations, defrost or safety conditions, and standby losses are not fully accounted for in the model. Additionally, the auxiliary energy consumed by circulation pumps, control systems, and measurement systems is not included in the calculated annual coefficient of performance. The actual electricity demand of the entire system may therefore be higher than calculated in the model.

Another limitation of the model lies in the constant heat output assumed for the circulation heat pump. The heat pump is modeled as having a maximum heat output of 500 W to the hot water storage tank. However, a real heat pump exhibits varying performance depending on source temperature, sink temperature, flow rate, compressor operation, and operating conditions. The model is therefore suitable for energy estimates but does not reflect the detailed performance and control behavior of the heat pump.

Performance-dependent modeling would, in principle, be feasible. The existing approach using the Carnot efficiency can serve as a basis for this, since the theoretical COP already depends on the temperature difference between the source and sink. However, for a more realistic representation, the heat output and electrical power of the heat pump would also need to be described as functions of source and sink temperatures. This would require either manufacturer performance charts, measurement data from reproducible tests, or a calibrated simplified performance chart model. This data was not available within the scope of this work.

The model includes further simplifications. The hot water storage tank is modeled only in terms of energy and not by temperature layers. As a result, mixing, inflow behavior, and real-world temperature stratification cannot be assessed in detail. The hydraulic conditions in the cold-water and hot-water networks are also accounted for in a simplified manner. Pressure losses, pump characteristics, valve settings, and possible circulation failures are not calculated dynamically.

In addition, ambient temperatures, circulation losses, and user behavior are modeled in a simplified manner. Circulation losses are assumed to be constant throughout the year, although in reality they depend on the temperature difference between the heated pipe and the surrounding environment. Room temperatures in shafts, utility rooms, or installation zones are not accounted for as time-dependent in the model. Hot water demand is also not varied seasonally, although seasonal differences can occur in real residential buildings. These simplifications particularly affect the calculated cold water temperature, the available storage capacity, and the operating times of the heat pump.

The most important model limitations and their impact on the results are summarized in Figure 26. The figure shows which simplifications are made in the model and which results are particularly affected by them. This makes it clear that the simulation should be understood as a potential assessment and does not replace a fully validated operational forecast.

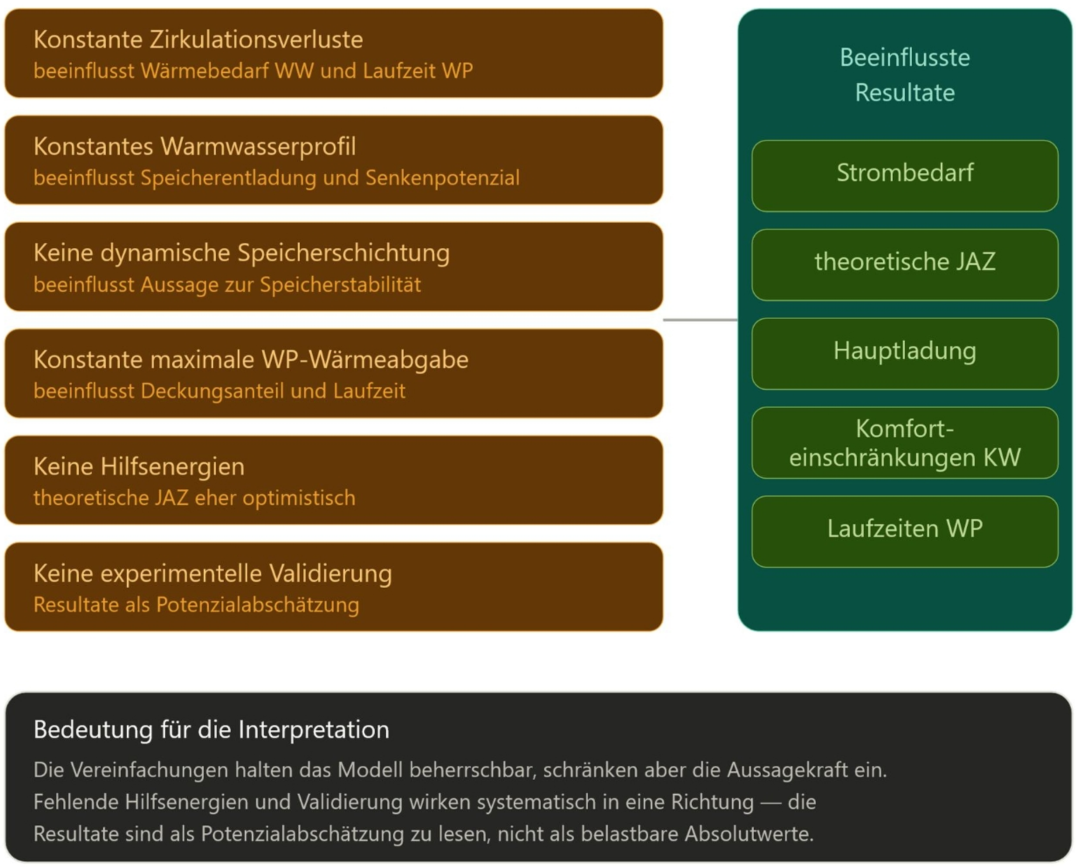

Figure 26 Model Limitations and Their Impact on the Results

It should also be noted that the model results could not be experimentally validated due to the failure of the heat pump. The simulation therefore does not provide a definitive energy assessment, but rather a computational estimate of potential under defined boundary conditions. A robust assessment would require reproducible measurement series with electrical power measurements, defined volume flows, and stable operating conditions.

For future work, the model should be calibrated using measurement data from reproducible experiments. Of particular importance for this are electrical power measurements of the heat pump, longer measurement series, and defined operating conditions involving hot and cold water withdrawal. Additionally, it would be useful to extend the model to include dynamic storage stratification in order to investigate the influence of heat input on storage stability more precisely.

In addition to the base case, a simplified sensitivity analysis was conducted. The goal of this analysis is not a comprehensive parameter study, but rather to estimate which input variables have a particularly strong influence on the model results. To this end, selected parameters are examined individually, while the remaining boundary conditions remain unchanged. The focus is on the cold-water temperature, the circulation losses in the hot-water distribution system, the hot-water demand, the available storage capacity, and the maximum heat output of the heat pump to the hot-water storage tank.

The cold-water temperature has a major influence, as it determines both the source potential and the theoretical COP. Higher cold-water temperatures improve the heat pump’s operating point and simultaneously increase the benefits of cold-water cooling. Lower cold-water temperatures, on the other hand, reduce the usable source potential and increase the temperature difference to the hot-water side. This reduces theoretical efficiency, and operation may be limited more quickly by the lower cold-water limit.

Circulation losses and available storage capacity are also relevant. Higher circulation losses increase the heat demand on the hot water side and can result in longer heat pump runtime. At the same time, however, this also increases the system’s total supplementary heating requirement. Conversely, a low hot water demand or a small storage tank limits the possible heat absorption on the sink side. As a result, the heat pump may be shut down despite the presence of heat in the cold-water network because the hot-water storage tank cannot absorb any additional heat.

The heat pump’s maximum heat output influences the extent to which it can cover circulation losses. However, higher capacity only results in an energy benefit if there is simultaneously sufficient source energy in the cold-water network and enough storage capacity on the hot-water side. If either of these two conditions is not met, higher heat pump capacity can lead to shorter runtime or more frequent shutdowns.

The sensitivity analysis serves to qualitatively classify the model’s most important influencing factors. Figure 27 shows the key parameters that influence the operation of the circulation heat pump. The focus is on the interaction between the source, the sink, and the heat pump. The heat pump can only be operated effectively if usable heat is available on the source side and heat can be absorbed on the sink side.

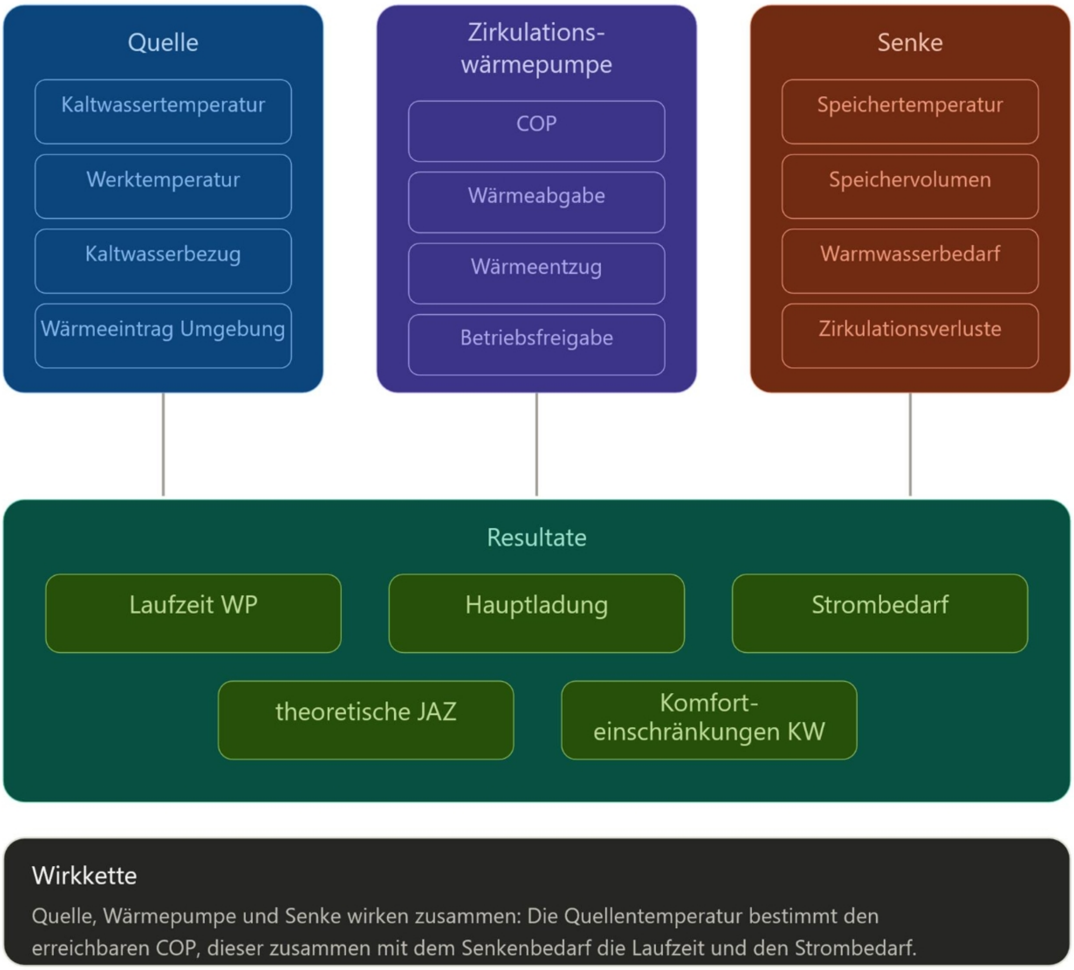

Figure 27 Factors Influencing the Operation of the Circulation Heat Pump

The most important influencing factors are summarized qualitatively in Table 19. The table serves as an overview of the expected effects of individual model parameters and does not replace a computational sensitivity analysis.

Table 19 Qualitative sensitivity analysis of the most important model parameters

Parameter	Change	Expected Effect
KW temperature	Higher	Better COP, higher cooling capacity, greater cooling benefit
KW temperature	lower	lower source potential, higher temperature difference, lower efficiency
Circulation losses	higher	Higher heat demand, potentially longer heat pump runtime
DHW Demand	higher	greater storage discharge, greater potential for reduction
DHW demand	lower	less storage discharge, more frequent shutdowns possible
Storage volume	larger	More buffer, longer runtime possible
Heat pump heat output	Higher	Higher coverage percentage possible, provided the source and storage are available
Ambient temperature	Higher	Higher heat input into the chilled water network, greater source potential

A computational sensitivity analysis with multiple scenarios would be useful for future work. For example, the cold water temperature could be varied by ±2 K, circulation losses by ±20%, hot water demand by ±20%, and the heat pump's maximum heat output. This would allow for quantifying which parameters most strongly influence the annual performance factor, the primary load, and the number of hours above and below the comfort range of the cold water.

In summary, the model shows that a circulation heat pump can contribute to the energy efficiency of the hot water storage tank. However, practical feasibility depends largely on the cold water temperature, storage management, control strategy, hydraulic stability, and the actual efficiency of the heat pump. The model thus provides a suitable basis for assessing potential; however, for a robust design, it must be validated with measurement data and expanded to include more detailed operating and control variables.

6 Discussion of the Results

6.1 Cold Water Network

The design of the cold water circulation system, as well as the fundamental hydraulic and technical relationships, were already described in Chapter 2.3.9 based on the previous work in the module
“Experimental Work.” There, a modular and mobile test setup for chilled water circulation was developed, constructed, and documented. The following discussion builds on these fundamentals and applies them to the planning and implementation of chilled water circulation in new and existing buildings.

The installation of a cold-water recirculation system places high demands on planning in both new and existing buildings. New and existing buildings must be considered differently, as the structural conditions, existing piping layouts, and technical possibilities can vary significantly. The goal of a cold-water recirculation system is to reduce unwanted temperature increases in the cold-water distribution system, prevent stagnation, and thereby improve drinking water hygiene. During planning, structural, hydraulic, microbiological, and operational constraints must be considered together.

When installing the circulation system in a pipe-to-pipe configuration, care must be taken to ensure that the pipe bundle is insulated tightly and airtight to prevent air pockets from forming. Due to the low medium temperatures, condensation may occur. For this reason, a conventional system is recommended for cold-water circulation. Figure 28 below shows a section of a piping diagram illustrating how cold-water circulation can be implemented at the floor level. Additionally, Figure 29 shows a schematic representation of the central plant. A complete piping diagram for the representative building is shown in Appendix B.2.

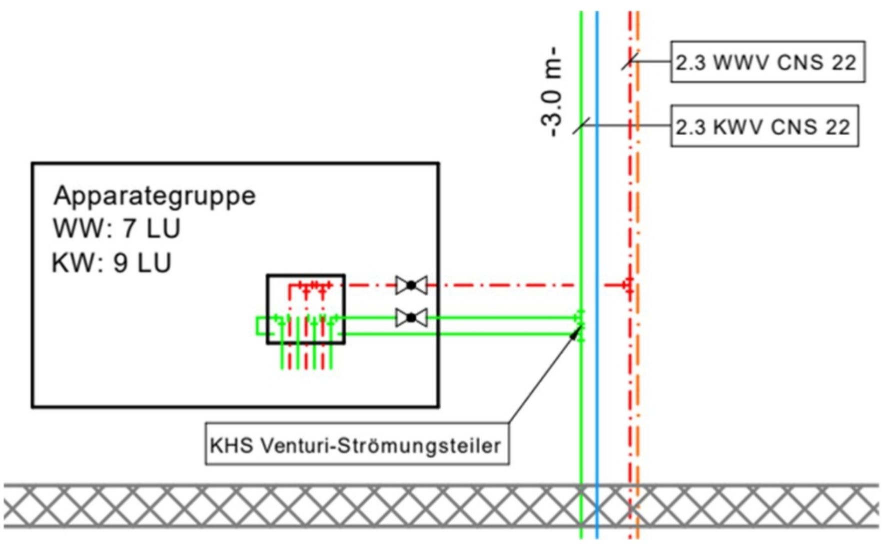

Figure 28 Floor-level distribution with KHS Venturi flow divider

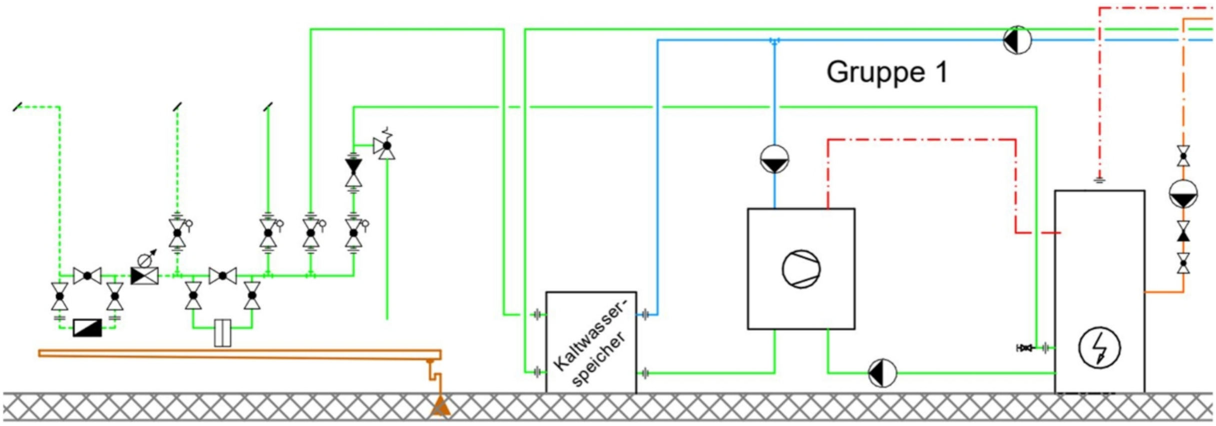

Figure 29 Schematic representation of the plumbing distribution

New Construction

A new construction project has the advantage that cold-water circulation can be taken into account as early as the planning phase. This allows utility rooms, riser zones, pipe routing, storage tanks, and any metering points to be specifically coordinated with one another. Pipe routing should be planned in such a way as to avoid unnecessary pipe lengths, dead-end pipes, and stagnation zones.

From a structural standpoint, care must be taken in new construction to ensure that sufficient space is available for the additional components. These include, among other things, cold-water circulation pipes, shut-off valves, backflow preventers, control valves, any flow meters and temperature sensors, as well as a storage tank or buffer tank for hydraulic decoupling. Likewise, accessibility for maintenance and operation must be ensured as early as the planning phase.

Hydraulically, a clear separation between the cold-water, hot-water, and heat pump circuits is required. Loop systems with Venturi flow dividers help ensure regular water exchange and prevent stagnation in the pipes (Eichelberger & Paganini, 2025; Kemper, n.d.-a). Measurement points for temperature, pressure, and flow rate are useful for monitoring operation and detecting circulation problems early on.

From a microbiological perspective, particular attention must be paid in new construction to ensuring that cold-water pipes are protected from excessive heat input so that microbiologically non-critical temperatures are not exceeded. This applies in particular to the spatial separation from hot pipes, appropriate insulation, and short pipe runs to the draw-off points. The goal is to avoid hygienically critical cold-water temperatures and to ensure regular water exchange.

Existing Buildings

In existing buildings, planning a cold-water circulation system is significantly more challenging, as the existing building structure and the existing drinking water system severely limit the options.

From a structural standpoint, particular attention must be paid to existing shafts, utility rooms, ceiling and wall penetrations, and fire safety requirements. It is often difficult to retrofit additional piping in existing buildings. Therefore, it must be determined whether existing pipe sections can be reused. A review of the existing pipe dimensions is required. In existing buildings, it is recommended to use a pipe-in-pipe system for cold-water circulation to counteract any lack of space. However, special care must be taken to prevent condensation, as moisture can form on the pipes due to the low fluid temperatures. As an alternative, a pipe-in-pipe circulation system can be considered. However, it should be noted that heat loss or heat gain is lower compared to other systems. This reduces the heat absorption by the cold water, causing the source temperature to drop. This study does not provide a definitive assessment of the thermodynamic effects this has on heat pump operation. In particular, it is unclear to what extent the lower source temperature of the cold water network affects the efficiency and operating mode of the heat pump.

From a microbiological perspective, the focus is particularly on assessing existing stagnation zones within the system. Rarely used draw-off points, long individual supply lines, or poorly routed pipes can lead to elevated temperatures and hygiene risks.

6.2 Hot Water System

The integration of a circulation heat pump particularly influences the thermal behavior of the hot water storage tank as well as storage management. The system's objective is not to fully heat domestic hot water, but rather to partially offset circulation heat losses by utilizing heat from the cold water network.

The studies show that the heat pump can help stabilize the upper zone of the storage tank. During the test measurements, it was observed that the cooling of the storage tank slowed down when the heat pump was active. At the same time, the storage tank exhibited stable stratification behavior without any noticeable significant mixing.

Heat is introduced specifically into the upper storage zone. This supports, in particular, the thermal stability of the top layer. This helps reduce the number of main charges required for the storage tank. At the same time, the existing hydraulic conditions within the hot water circulation system remain unchanged.

In addition, the concept offers the possibility of selectively thermally overloading the hot water storage tank. This appears particularly useful during the summer months, as higher temperatures occur within the cold water network during this time, making more usable thermal energy available. Especially in buildings with low hot water consumption, this allows for better utilization of the heat pump's energy potential.

For existing systems, the concept offers additional advantages, as no significant adjustments to the existing hot water circulation are necessary. However, it remains a prerequisite that the circulation functions hydraulically correctly and has been properly adjusted.

6.3 Integration of a Circulati e Heat Pump

Various hydraulic integration concepts were evaluated for the implementation of the circulation heat pump. The goal was a solution that could operate with hydraulic stability, have minimal impact on the existing hot water circulation, and be retrofittable in both new construction and existing buildings.

The evaluation of the different variants is based on the requirements for hot water circulation, storage tank management, and hydraulic balancing described in Chapter 2. In addition, the findings from the test bench were taken into account. The focus was on stable flow rates on the load side of the heat pump, minimizing the impact on the existing hot water circulation, and ensuring controlled heat transfer into the hot water storage tank.

To better contextualize the variants under consideration, Figure 30 illustrates the principle of direct integration into the circulation return line. This variant takes the approach of reheating circulation losses directly where they occur in the system.

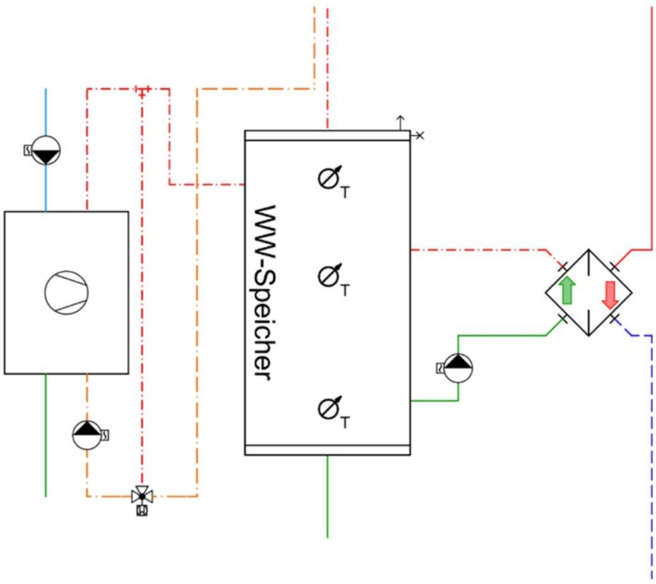

Figure 30: Principle of direct integration into the circulation return

With a direct integration into the hot water return or circulation return, the cooled circulation water would be reheated directly via the heat pump and then returned to the hot water distribution system or to the storage tank. The advantage of this solution lies in the direct compensation for circulation heat losses. At the same time, the heat would be introduced directly into the affected water circuit.

However, this option was not pursued further because it presents various operational disadvantages for a small circulation heat pump. The hot water circulation system is designed primarily to meet hygienic and hydraulic requirements, rather than the operational requirements of a heat pump. The existing flow rates, temperature differentials, and pressure conditions therefore do not necessarily meet the heat pump’s minimum requirements. In particular, the required minimum flow rate on the load side represents a key boundary condition. Direct integration is therefore considered hydraulically critical, as stable operation of the heat pump cannot be guaranteed under all operating conditions.

In addition, interactions may arise between the existing circulation control system and the heat pump’s operation. Thermal control valves, varying pipe lengths, or changing operating conditions can cause the flow rates within the circulation system to fluctuate. If the heat pump is integrated directly into this circuit, this can result in unstable operating conditions, frequent cycling, or insufficient heat output. Furthermore, interfering with the hot water circulation would complicate the design, commissioning, and subsequent maintenance of the existing system.

As a further alternative, integration with an additional buffer tank in the circulation return line was considered. This solution could hydraulically stabilize the heat pump’s flow rate, as the heat pump would not be directly dependent on the current circulation flow rate. At the same time, however, an additional water-carrying storage tank would be introduced into the domestic hot water system. This increases space requirements, control complexity, and hygiene requirements. Due to these disadvantages, this variant was not experimentally implemented as part of this study and was not explored further.

To illustrate the preferred solution, Figure 31 shows the hydraulically decoupled integration via the hot water storage tank. In this configuration, the heat pump is not directly integrated into the hot water circulation but is coupled to the storage tank via a separate heat pump circuit.

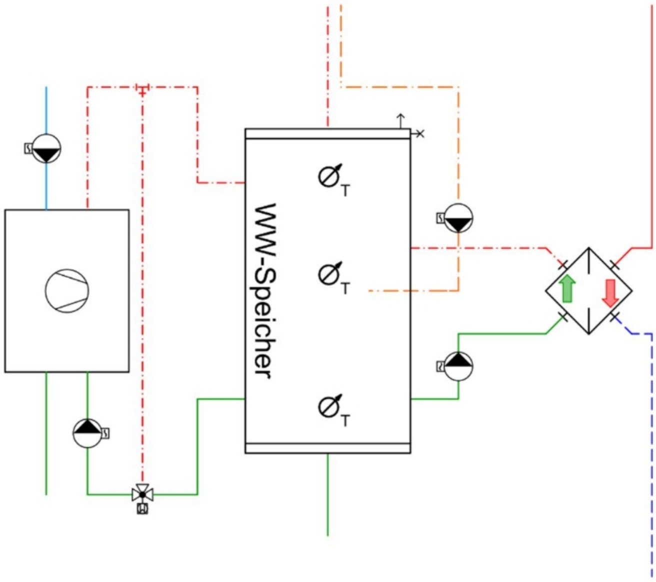

Figure 31 Hydraulically Decoupled Integration

For this study, the hydraulically decoupled integration via the hot water storage tank is preferred. The heat pump draws water from the storage tank, heats it, and then returns it to the upper section of the tank. This creates a separate heat pump circuit that operates in parallel with the existing hot water circulation system. The hot water circulation system remains hydraulically independent and can continue to be planned, sized, and adjusted according to standard practices.

The main advantage of this solution lies in the more stable operating conditions for the heat pump. The flow rate in the heat pump circuit can be adjusted independently of the circulation flow rate. This allows for more precise control of minimum flow rates, temperature differentials, and return temperatures. At the same time, it reduces the risk of the heat pump being affected by fluctuating circulation conditions.

This type of integration is also advantageous for existing systems. Provided that suitable storage tank connections are available or can be retrofitted, the existing hot water circulation system does not need to be significantly modified. This reduces the planning and installation work required for the existing drinking water system. However, it remains a prerequisite that the existing circulation system functions hydraulically correctly and is properly adjusted.

Direct integration into the hot water supply line was not pursued further. The hot water supply line is already at the temperature required to serve the draw-off points. Additional heating at this point would not specifically compensate for circulation losses and could lead to unnecessarily high system temperatures. The hot water return line—or circulation return line—is more obvious from an energy perspective, but was not selected as the preferred option due to the hydraulic interactions described above.

Installing the heat pump directly between the cold-water line and the hot-water line is also viewed critically. The cold-water side can be used as a heat source, but must

They must remain hygienically and hydraulically separated from the hot water system. Furthermore, the cold drinking water system and the hot drinking water system must not be mixed directly. Energy coupling is therefore only practical via separate heat exchangers or via the heat pump’s refrigeration circuit. The cold water line thus serves as the source, while the hot water storage tank is the preferred sink.

Hydraulic integration via the storage tank also influences thermal stratification. Targeted injection into the upper zone of the storage tank can support the top layer. At the same time, it is essential to prevent excessively high flow rates or unfavorable inflow conditions from causing the storage tank to mix. For this reason, maintaining a high return flow temperature, the inflow position, and flow rate control are key factors for further optimizing the system.

The most important differences among the connection types considered are summarized in Table 20. The table serves as a qualitative evaluation of the variants and shows why connecting the storage tank is preferred over direct integration into the circulation system.

Table 20 Evaluation of Connection Types

Variant	Main Advantage	Assessment
WWR in circulation return	Direct reheating	Hydraulically critical
WWR with buffer tank	More stable heat pump flow rate	Hygienically and technically complex
Storage tank connection	Decoupled heat pump circuit	Preferred solution

Direct integration into the domestic hot water return or circulation return allows for immediate reheating of circulation losses. However, this variant is heavily dependent on the existing circulation flow rate. This can lead to interactions with the circulation control system, which means that the heat pump’s minimum flow rate cannot be guaranteed under all operating conditions.

Integration with an additional buffer tank in the circulation return could hydraulically stabilize the heat pump’s flow rate. However, this would introduce an additional water-carrying storage tank into the domestic hot water system. This increases space requirements, the complexity of the control system, and hygiene requirements. This variant was therefore only considered conceptually and was not experimentally implemented within the scope of this study.

The evaluation of the options shows that the storage tank connection represents the most suitable solution for this study. It enables a hydraulically decoupled heat pump circuit, allowing the flow rate, temperature spread, and return temperature to be adjusted independently of the existing hot water circulation. At the same time, the domestic hot water circulation remains largely unchanged. This reduces the risk of interactions with circulation valves, fluctuating branch flow rates, or insufficient minimum flow rates for the heat pump.

6.4 Requirements n Circulation Heat Pumps

The investigations in this study identify requirements for a future circulation heat pump. The focus is not only on thermodynamic operation but, in particular, on drinking water hygiene, hydraulic stability, operational safety, and easy integration into existing drinking water systems.

For a future market-ready circulation heat pump, requirements arise from several areas. In addition to the refrigeration circuit, hydraulic integration, drinking water hygiene, control systems, maintenance, and building integration are particularly crucial. Table 21 summarizes these requirements for a future circulation heat pump in tabular form. This overview is derived from the investigations in this thesis and simultaneously forms the basis for the specifications in Appendix D.1.

Table 21 Requirements for a Future Circulation Heat Pump

Area	Requirement
Operational reliability	Stable operation under varying source and sink temperatures, as well as with a large temperature difference between the cold water source and the hot water storage tank
Refrigeration cycle	Suitable compressor, safe refrigerant design, stable control of the refrigeration circuit, and consideration of the refrigerant charge, especially for R290
Hydraulics	Hydraulically decoupled heat pump circuit with a defined minimum flow rate independent of the existing hot water circulation
Drinking Water Hygiene	SVGW-compliant water-carrying components, no stagnation zones, no circulation failures, and no hygienically critical temperature ranges
Chilled Water Side	Use of the cold water network as the primary heat source while adhering to temperature limits and ensuring regular water exchange
Hot water side	Compliance with requirements for temperature control, supply reliability, and storage management
Heat transfer	Separation of the cooling circuit and the drinking water system via suitable heat exchangers; retrofitability and service accessibility must be taken into account
Control	Overarching system control that takes into account chilled water temperature, storage tank temperature, storage capacity, minimum flow rates, and possible alternative sources and sinks
Alternative sources/sinks	Expandability to accommodate additional sources or sinks—such as utility rooms—using appropriate switching valves and clear operating logic
Maintenance and Operation	Compact design, good accessibility, filtration, venting, shut-off options, and easy maintenance
Monitoring	Measurement of temperatures, airflow rates, and electrical power for monitoring as well as for determining the COP and annual performance factor
Building Integration	Easy integration into building automation systems, low noise emissions, and clear installation requirements

The table shows that a circulation heat pump cannot be viewed in isolation as a refrigeration unit. For practical implementation, the refrigeration circuit, drinking water hygiene, hydraulics, and control systems must be designed in conjunction with one another. Particularly critical factors here are the secure separation of drinking water systems, the prevention of stagnation, and stable operating conditions on both the source and sink sides.

For a market-ready implementation, it is particularly important that the heat pump can be operated hydraulically independent of the hot water circulation system. This allows for more precise control of minimum flow rates, temperature differentials, and return temperatures. At the same time, it prevents the existing circulation hydraulics from being negatively affected by heat pump operation.

From a hygienic standpoint, SVGW-compliant design is essential. All water-carrying components must be suitable for drinking water. Furthermore, the integration of the heat pump must not result in any additional stagnation zones or uncontrolled circulation failures. The thermal coupling between the cold water and hot water sides must therefore be achieved via suitable heat exchangers or through the refrigeration circuit.

The control of the circulation heat pump must go beyond the internal heat pump control system. A critical requirement is a higher-level system control that activates the heat pump only when usable heat is available on the source side and the hot water storage tank on the sink side is capable of absorbing heat. If the cold water is too cold or the storage tank is fully charged, the heat pump must be deactivated or switched to an alternative operating mode.

A simplified control strategy for different operating modes was derived from the requirements described. This is presented in Appendix D.2. In addition, the most important requirements for a future market-ready circulation heat pump were summarized in a specification document. This specification document serves as the basis for the potential industrialization of the system and is included in Appendix D.1.

7 Conclusion and Outlook

In summary, this study demonstrates that the hydraulic integration of a circulation heat pump into the cold and hot water networks is fundamentally feasible. The cold water network can be used as a heat source to cover a portion of the circulation heat losses in the hot water system. At the same time, heat can be extracted from the cold water, which can provide hygienic and comfort-related benefits, particularly when cold water temperatures are elevated.

Theoretical analyses show that the usable potential depends heavily on the location and the prevailing cold-water temperatures. In regions with low cold-water temperatures, the benefits are limited, whereas in urban areas with higher summer temperatures, there is greater potential. For the representative building, a heat loss from the hot-water distribution system was calculated at 484.5 W. In contrast, the potential heat absorption by the cold-water network is 329.7 W. The circulation heat losses cannot be fully covered by the cold-water network; however, a significant portion can be utilized.

For integrating the heat pump, direct integration into the return line of the hot-water circulation system proved to be hydraulically problematic. The fluctuating flow rates and potential interactions with the circulation control system make stable operation of the heat pump difficult. A more advantageous approach is a hydraulically decoupled integration via the hot-water storage tank. In this configuration, water is drawn from the storage tank, heated by the heat pump, and then returned to the upper section of the storage tank. This allows the hot-water circulation system to be designed, operated, and adjusted independently of the heat pump.

Commissioning and test measurements showed that the test setup functions hydraulically. The heat pump used in the laboratory, with a cooling capacity of approximately 600 W, was able to extract heat from the cold water system and transfer it to the hot water storage tank. Furthermore, the measurements indicated that maintaining a high return temperature on the load side supports stratification within the storage tank. However, after approximately 20 hours of operation, the heat pump—specifically the compressor—failed. As a result, it was not possible to complete a full set of measurement series.

A conclusive energy assessment is therefore not possible in this study. In particular, reliable conclusions regarding the actual COP, the annual performance factor, and the long-term behavior of the system cannot be drawn due to the limited data set. However, the study provides insights into the hydraulic feasibility, the operational behavior of the test setup, and the requirements for future systems.

The supplementary calculation model shows that the circulation heat pump can provide energy support to the hot-water storage tank. In the model, the heat pump transfers 4,380 kWh/a of heat to the hot-water storage tank. Of this, 2,947 kWh/a is extracted from the cold-water system. The calculated electricity demand is 1,434 kWh/a, resulting in a theoretical annual performance factor of approximately 3.06.

The main charge, at 11,066 kWh/a, remains necessary. The heat pump therefore does not replace the main heat generator but acts as a supplementary system for utilizing heat from the chilled water network. An analysis of the chilled water temperature also shows that the comfort upper limit of 15 °C is not exceeded in the model. At the same time, during cold periods, the temperature falls below the comfort lower limit of 8 °C. For future implementation, therefore, a temperature-dependent activation or power control is required to ensure that the chilled water is neither too warm nor unnecessarily cooled.

A comparison with direct electric reheating demonstrates the energy advantage of the heat pump. While the heat pump in the model requires 1,434 kWh/a of electricity, direct electric heating of the same amount of heat would require approximately 4,380 kWh/a of electricity. However, this comparison does not constitute a complete economic analysis. For an economic evaluation, additional investment costs, auxiliary energy for the circulation pumps, maintenance, servicing, service life, and actual operating times would also need to be taken into account.

From a hygiene perspective, SVGW-compliant design is particularly crucial for future implementation. All water-carrying components must be suitable for drinking water. Furthermore, the integration of the heat pump must not result in stagnation zones, circulation failures, or hygienically critical temperature ranges. This applies to both the cold-water system and the hot-water system.

For future work, it is recommended to first conduct a detailed investigation into the cause of the heat pump failure and to improve the system’s operational reliability. Subsequently, reproducible measurement series involving electrical power measurements should be carried out so that the efficiency of the circulation heat pump can be assessed. Furthermore, longer measurement series should be conducted with varying cold-water and hot-water draw-offs to examine the system’s operational behavior under real-world conditions.

In addition, studies on alternative heat sources and heat sinks, on storage tank overcharging, and on control strategies appear to be worthwhile. The design of the heat transfer between the refrigeration circuit and the drinking water system should also be examined in greater depth. In particular, it must be determined whether external heat exchangers or specially developed storage tanks with integrated heat exchangers are more suitable for a future market-ready implementation. Finally, the SVGW compliance of the entire system must be verified before industrialization can proceed.

The evaluated well temperatures also show that the potential of the circulation heat pump is highly dependent on location. While comparatively low cold water temperatures occur even in summer in alpine regions such as St. Moritz and Chur, significantly higher temperatures are found in the urban area of Zurich-Höngg. For future work, a geographic potential assessment would therefore be useful. This could involve comparing well temperatures, elevation, water source, settlement structure, and population distribution. Such an analysis would reveal which regions of Switzerland a circulation heat pump could be particularly relevant due to higher cold-water temperatures. However, a quantitative, Switzerland-wide potential map is not included in this study, as it would require additional temperature data from other water supply systems.

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Abbreviations

List of Units

a	Year
h	Hour
K	Kelvin
kWh	kilowatt-hour
l	Liter
m	meter
mm	millimeter
W	Watt

Greek letters

λ	Lambda
π	Pi

Abbreviations

COP Coefficient of Performance

FWS Freshwater Station

JAZ Annual Coefficient of Performance

KBE Colony-Forming Unit

KW Cold water

LU Loading Units

DHW Hot Water for

Drinking WP Heat

Pump WW Hot Water

WWR Hot Water Return

WWV Hot Water Supply

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AI- : Usage

AI-based tool	Use Case	Field	Remarks
ChatGPT (OpenAI) Last accessed: June 2, 2026	Revision of text regarding Conciseness, grammar, spelling, and style	Entire paper	-
Claude Pro (Anthropic) Last accessed: June 2, 2026	Support with code, diagrams, analyses, and the creation of the potential map	Entire work	-

Appendix